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Sitter et al.

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(54) **BELT SANDER**

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(52) **U.S. Cl.**
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Primary Examiner — Eric J Rosen

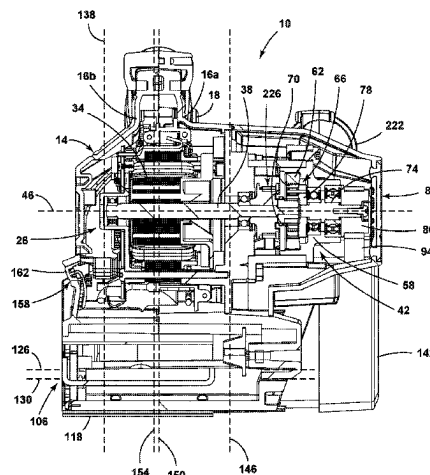
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(57) **ABSTRACT**

A belt sander for sanding a workpiece and defining a center of gravity within a first plane. The belt sander includes a housing, a handle defining a second plane, and a drive unit disposed within the housing. The drive unit includes a motor disposed adjacent a first sidewall of the housing, a pulley system disposed adjacent a second sidewall of the housing opposite the first sidewall, and a transmission disposed between the motor and the pulley system. The belt sander further includes a belt drive system driven by the drive unit and including a drive wheel, a driven wheel, and a platen. The platen defines a third plane extending perpendicular therefrom. The transmission and pulley system are disposed

(Continued)



on one side of the first plane and the motor is disposed on an opposite side of the first plane. The motor is intersected by the second and third planes.

17 Claims, 19 Drawing Sheets

Related U.S. Application Data

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(58) Field of Classification Search

CPC . B25F 5/026; B25F 5/02; B23D 51/01; B25G 1/102

See application file for complete search history.

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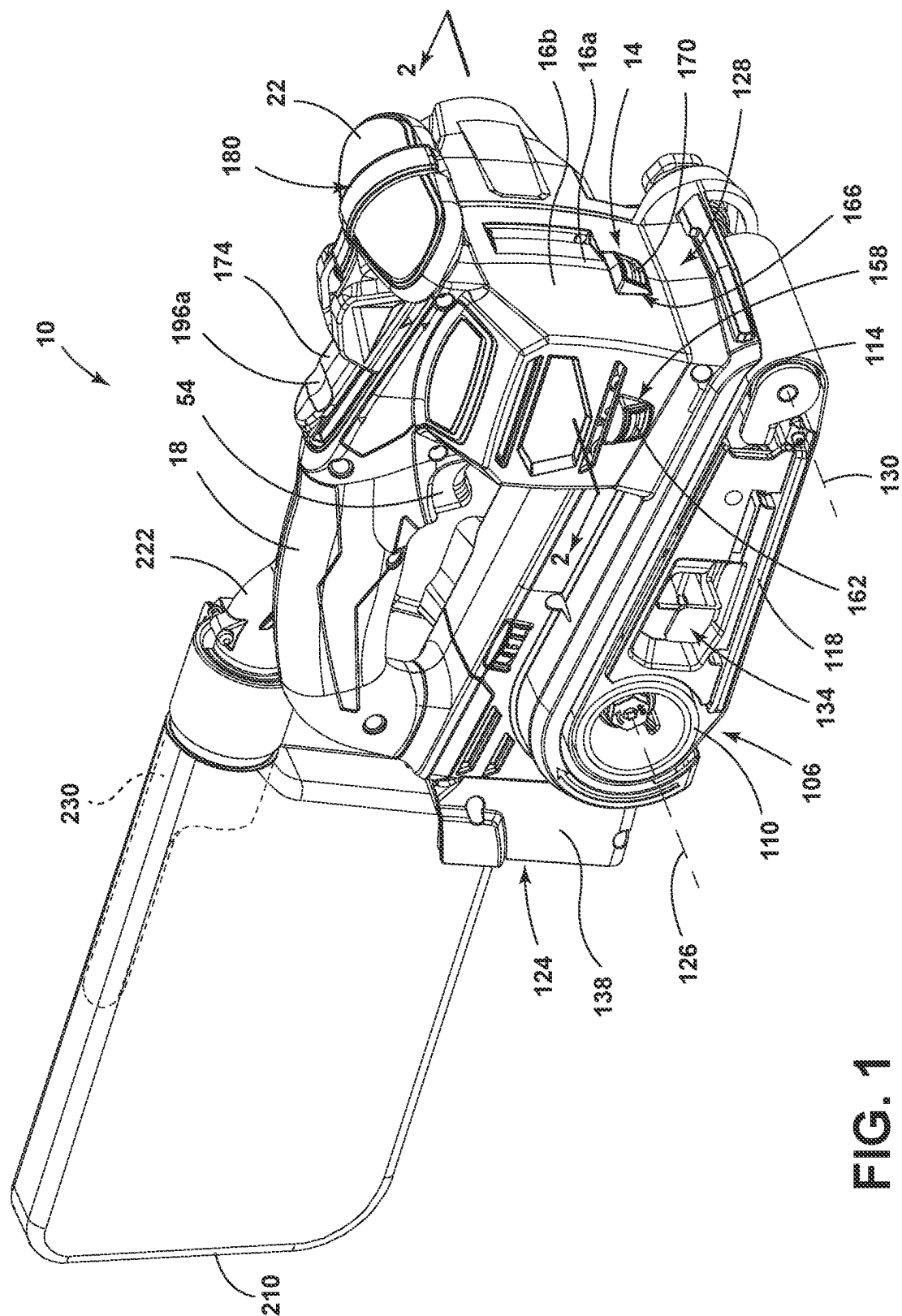
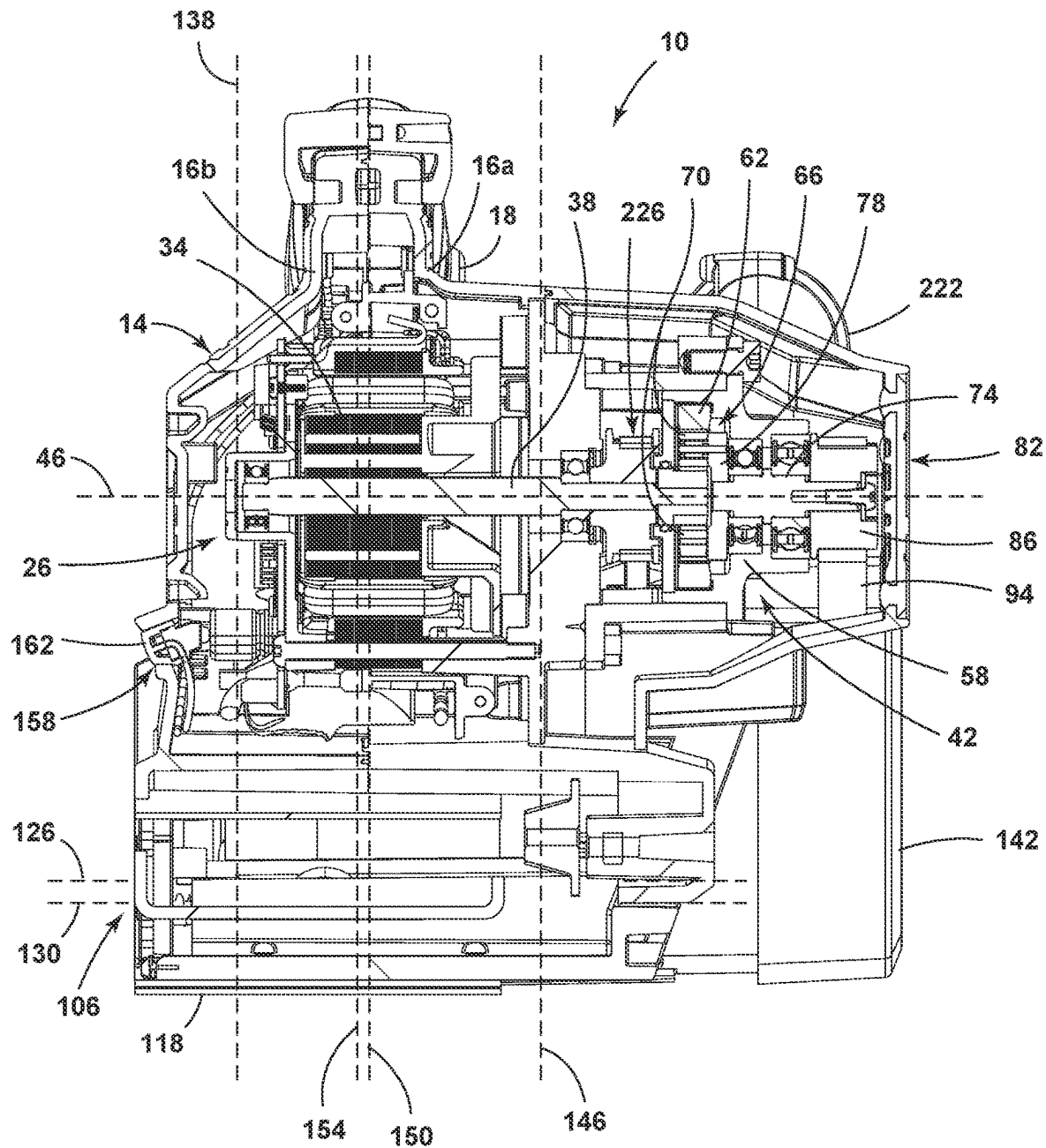


FIG. 1

**FIG. 2**

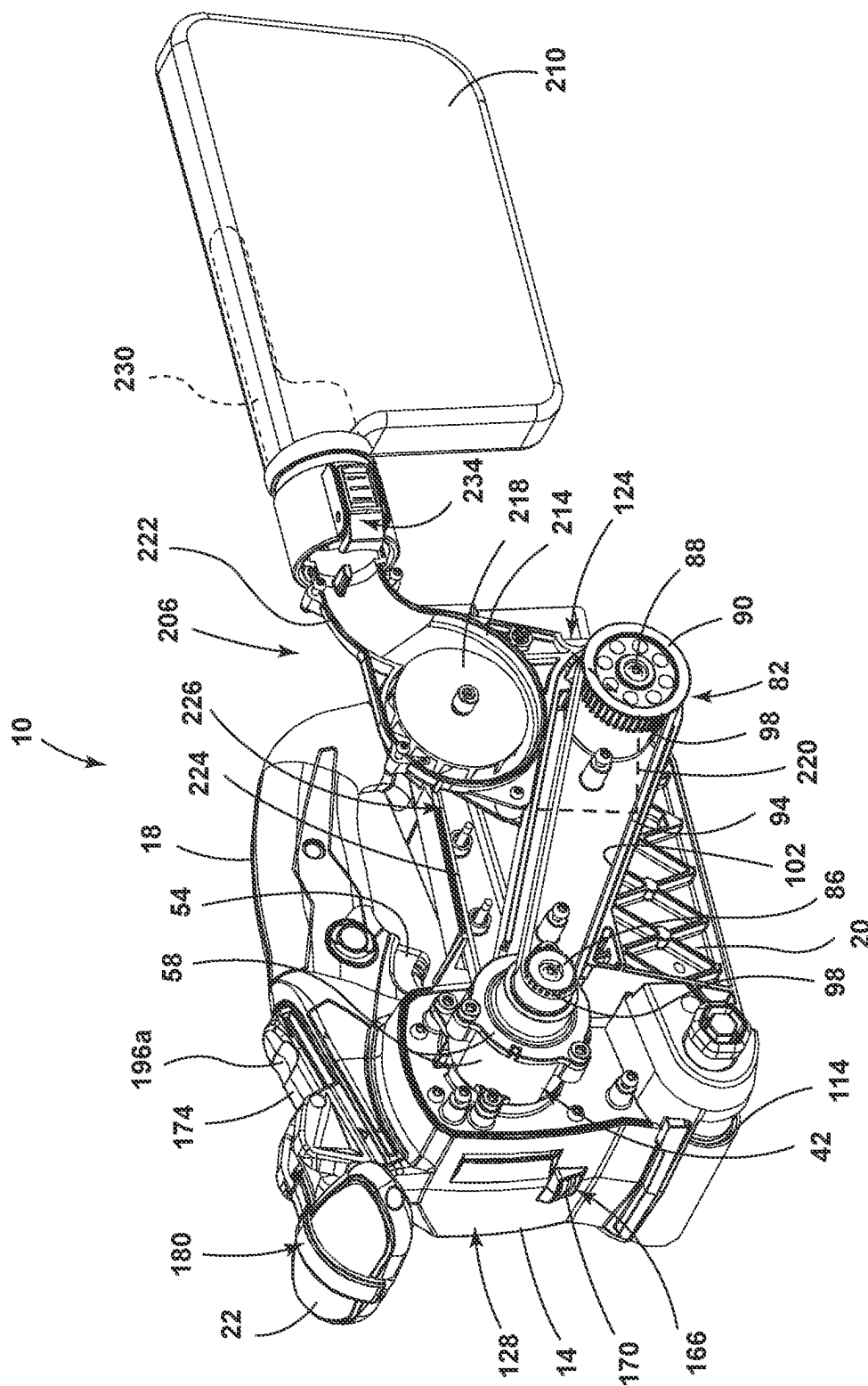


FIG. 3

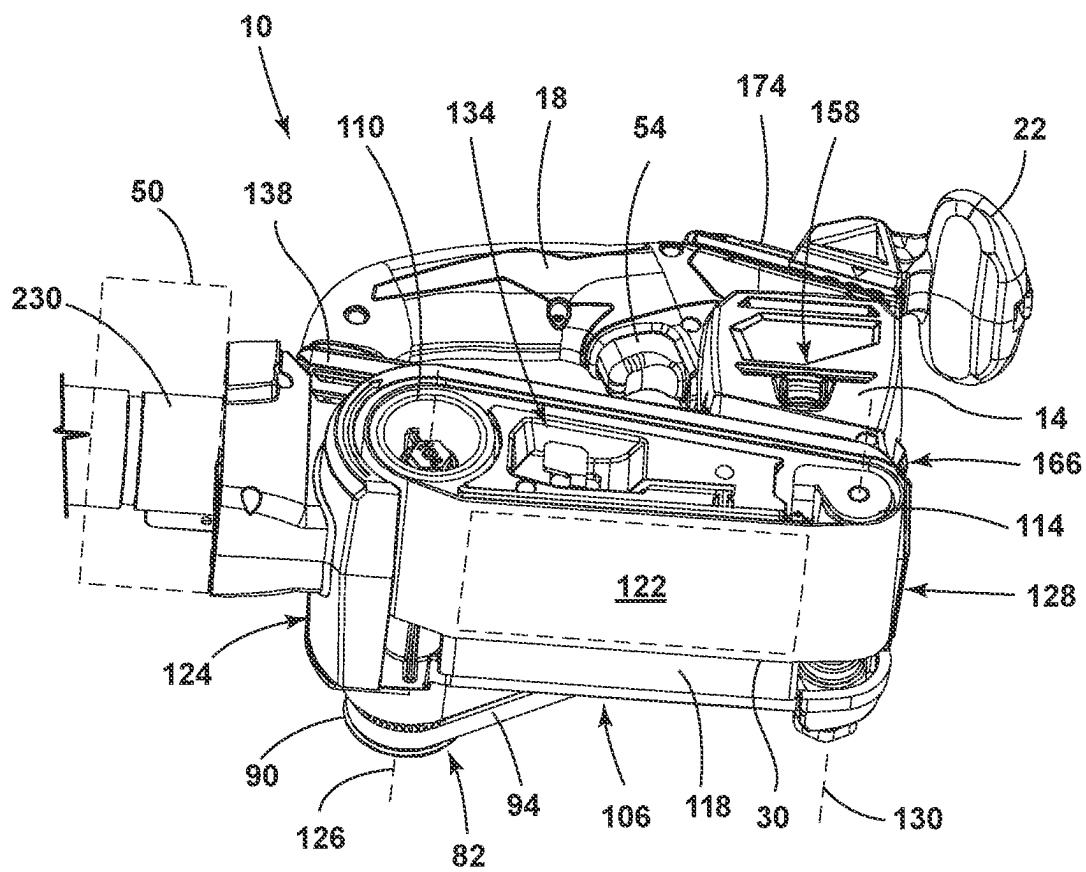
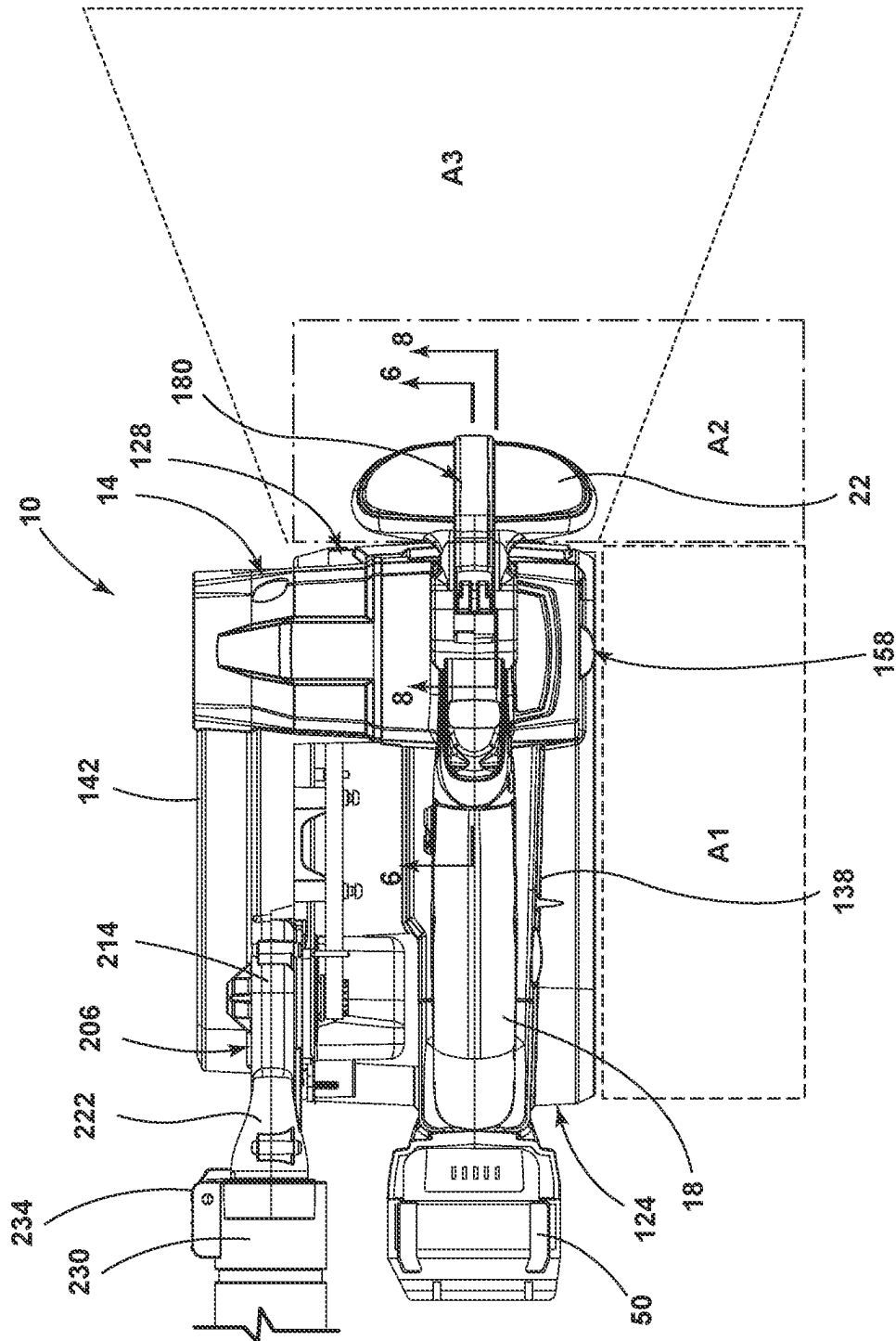


FIG. 4



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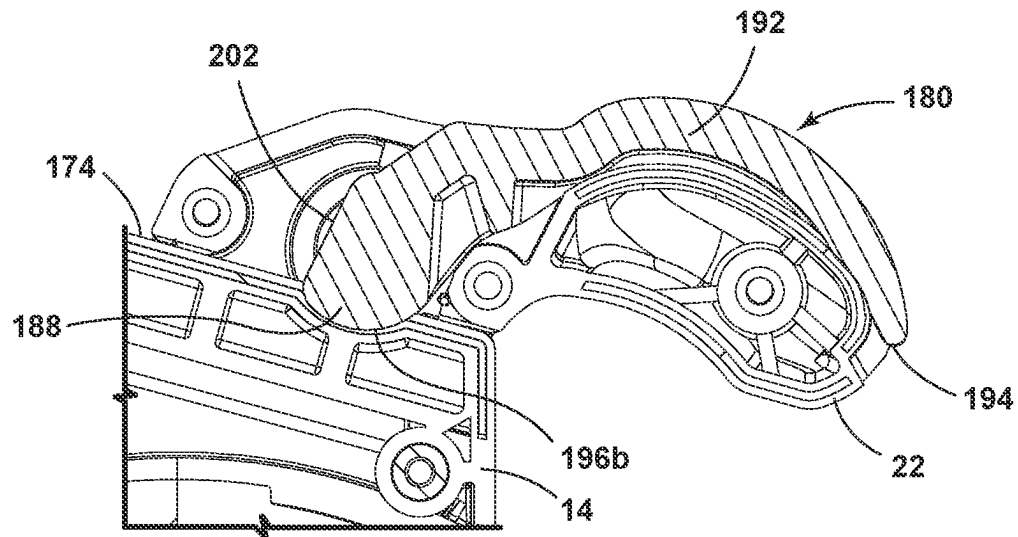


FIG. 6

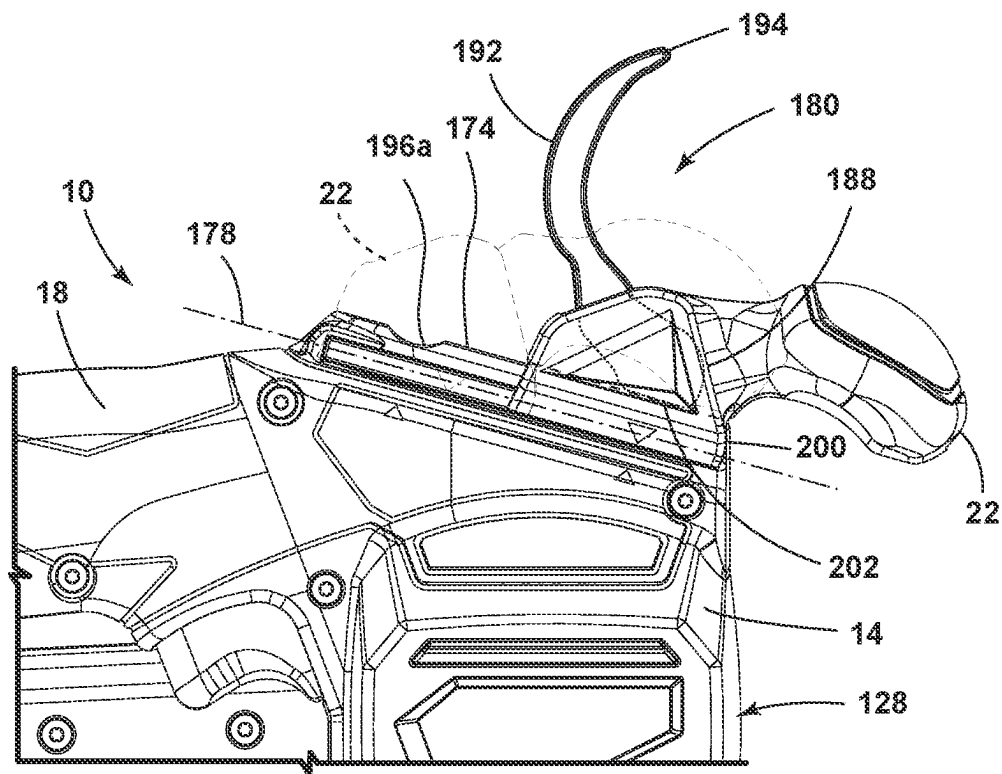


FIG. 7

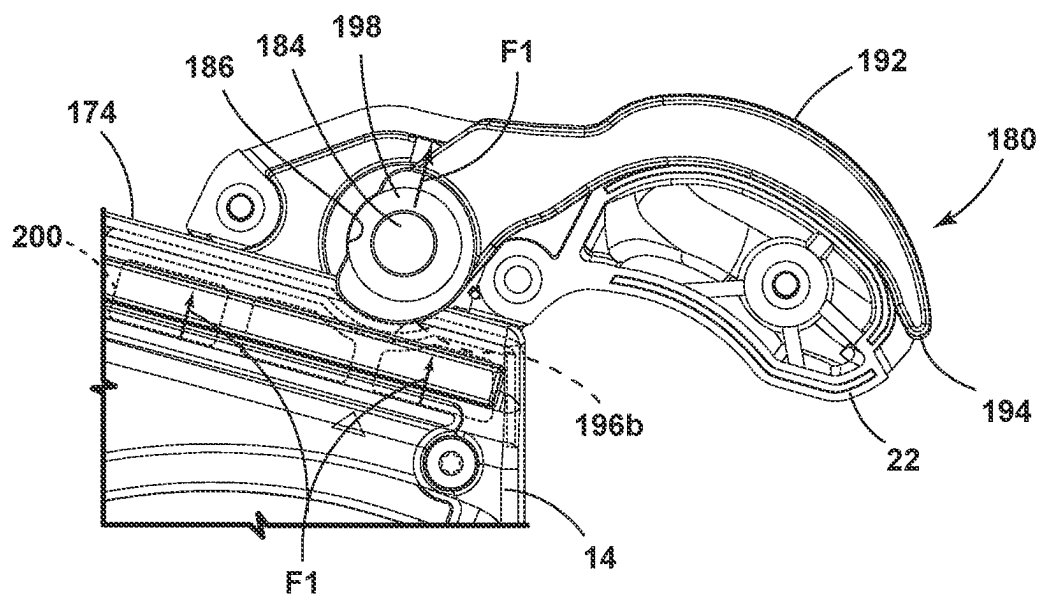


FIG. 8

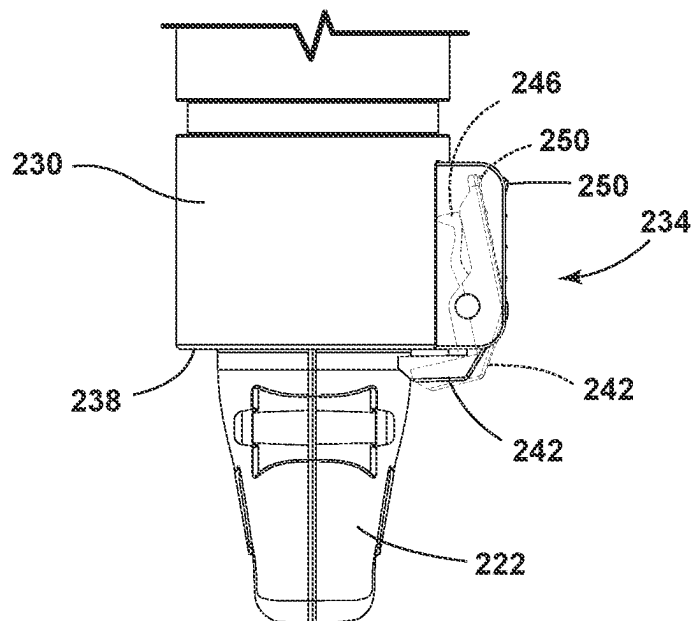


FIG. 9

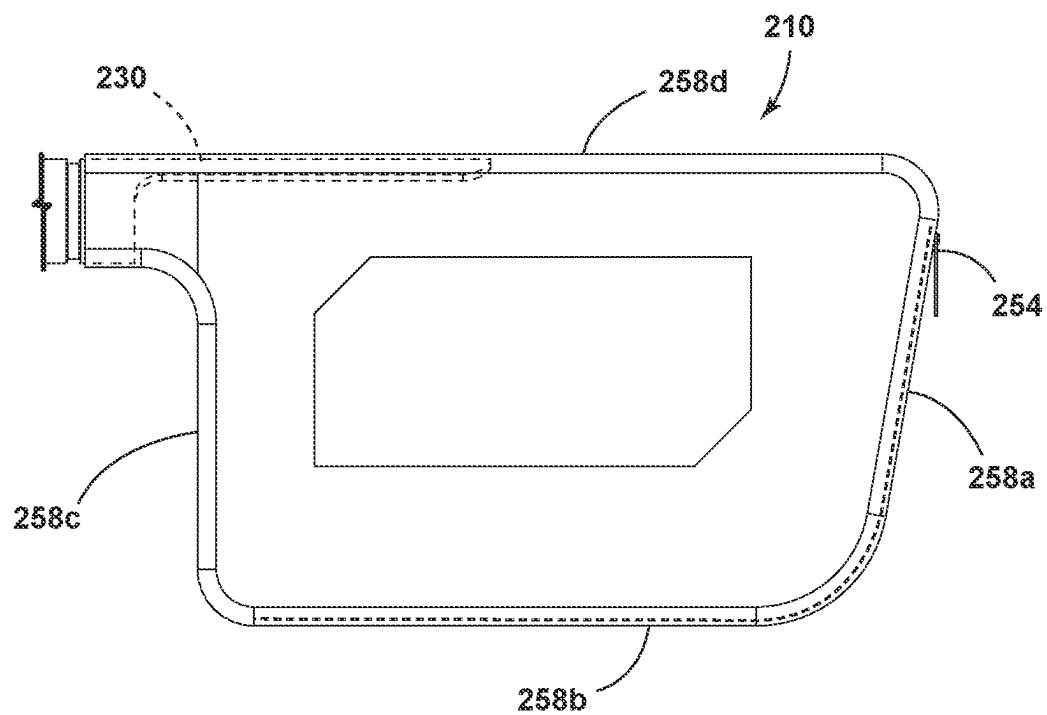


FIG. 10

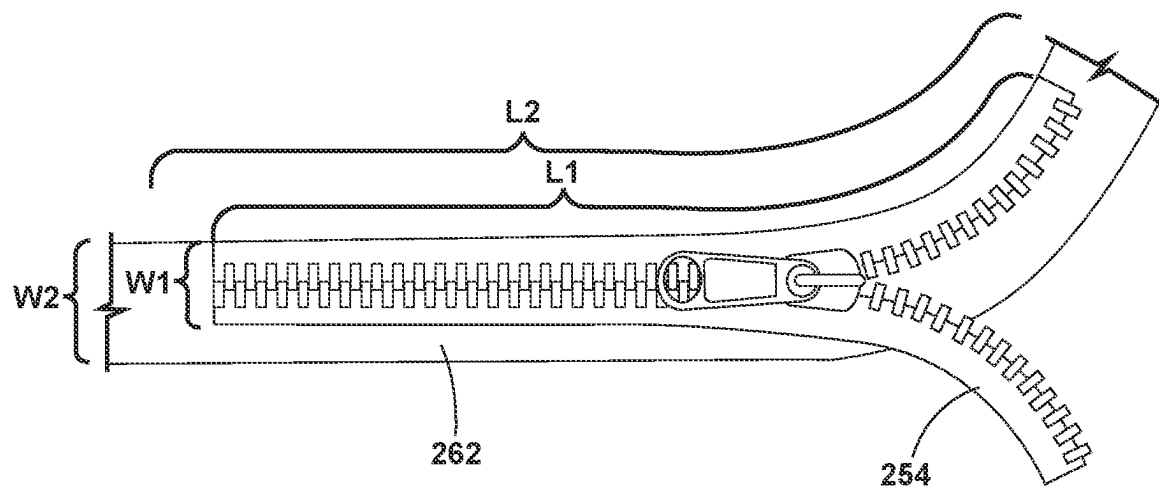


FIG. 11

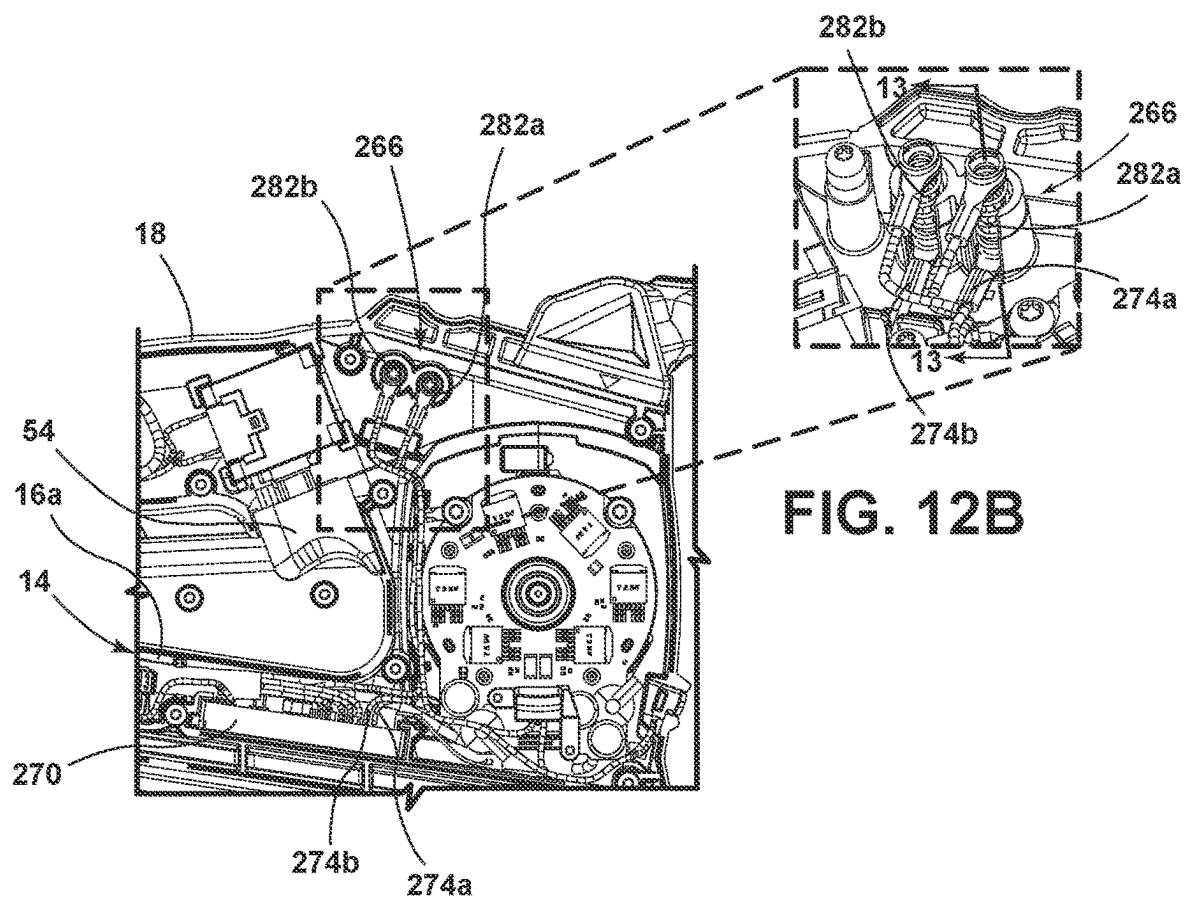


FIG. 12A

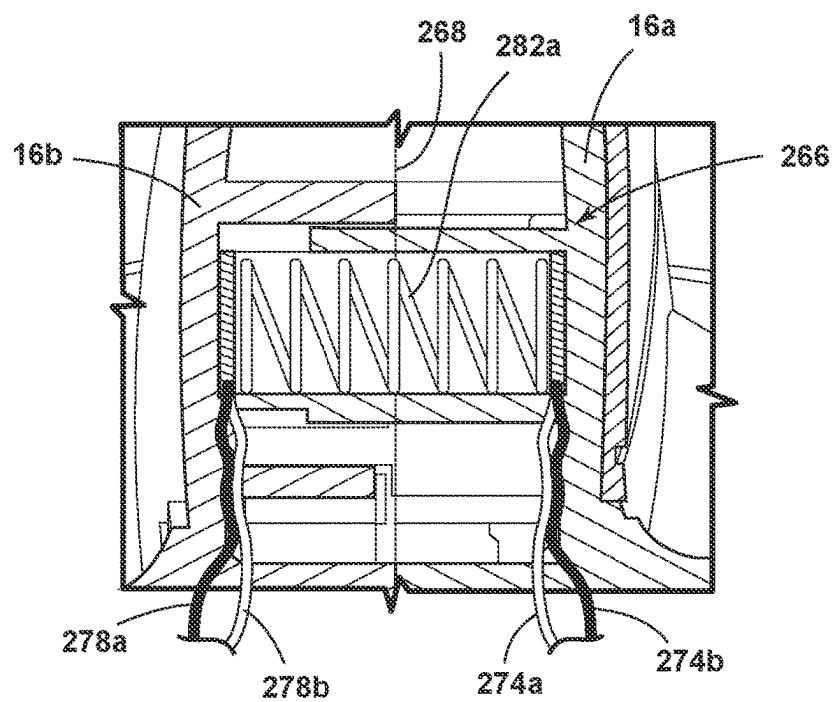


FIG. 13

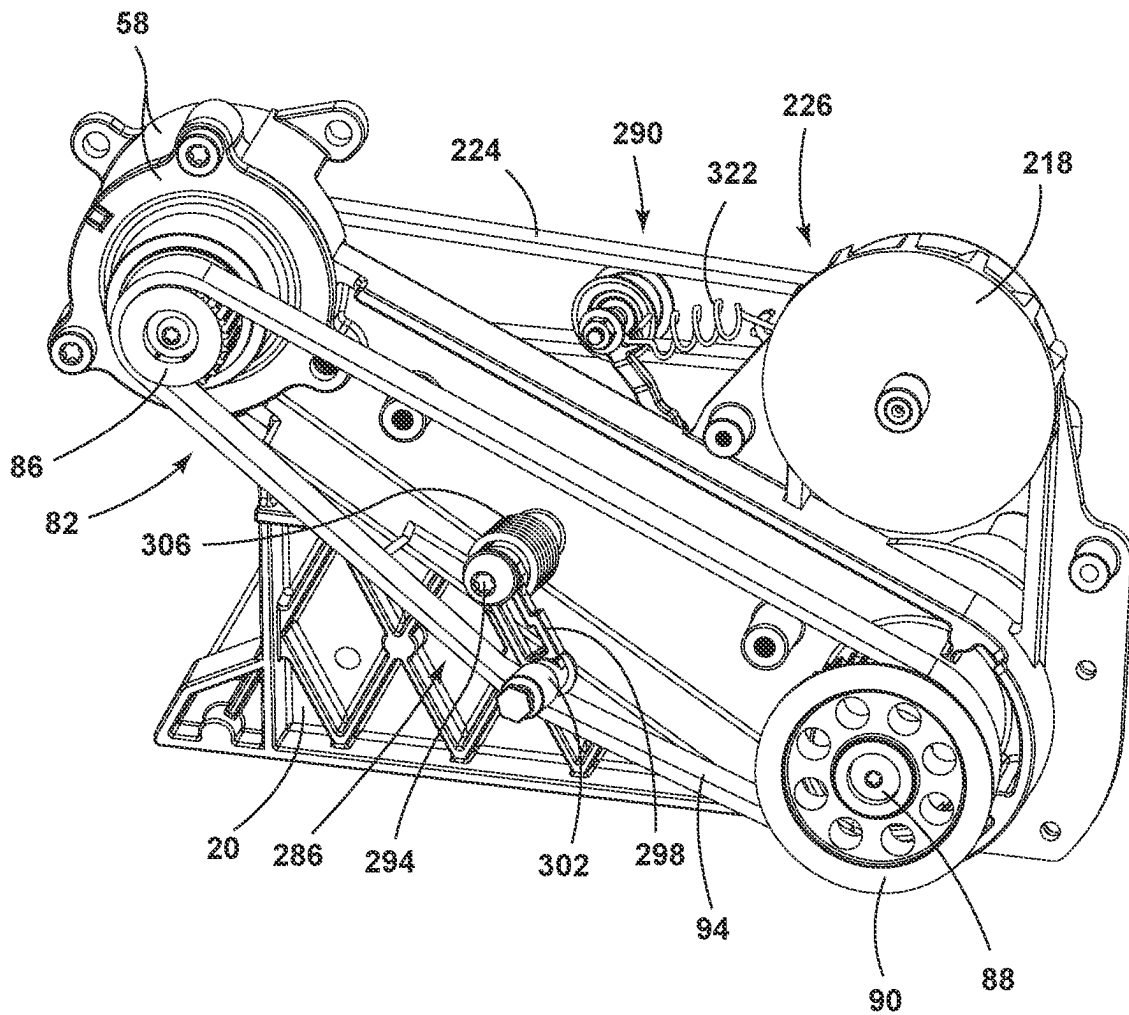
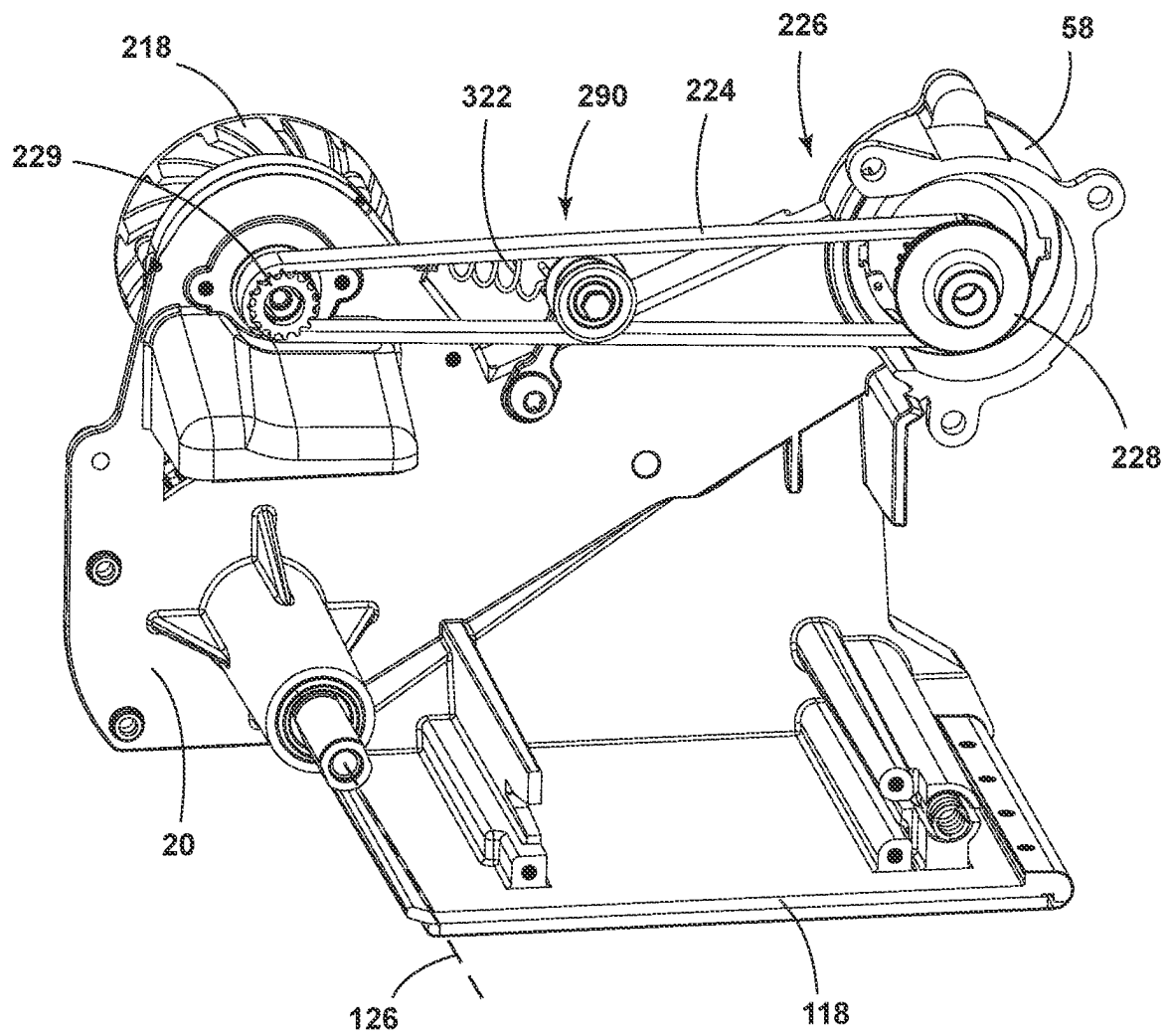


FIG. 14

**FIG. 15A**

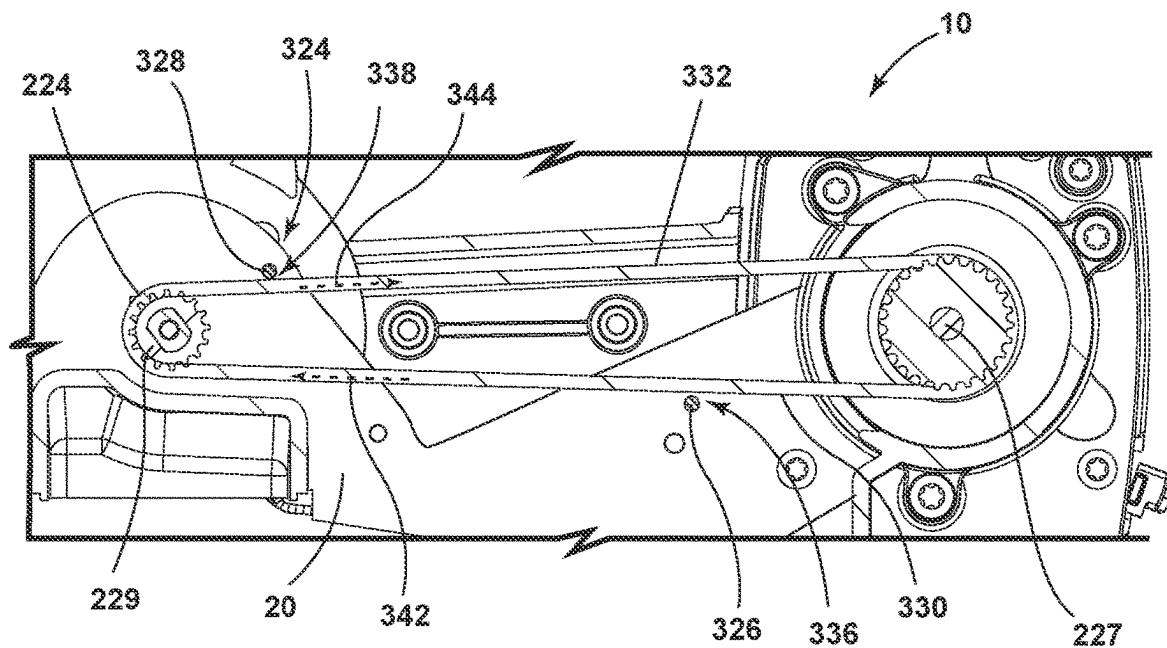


FIG. 15B

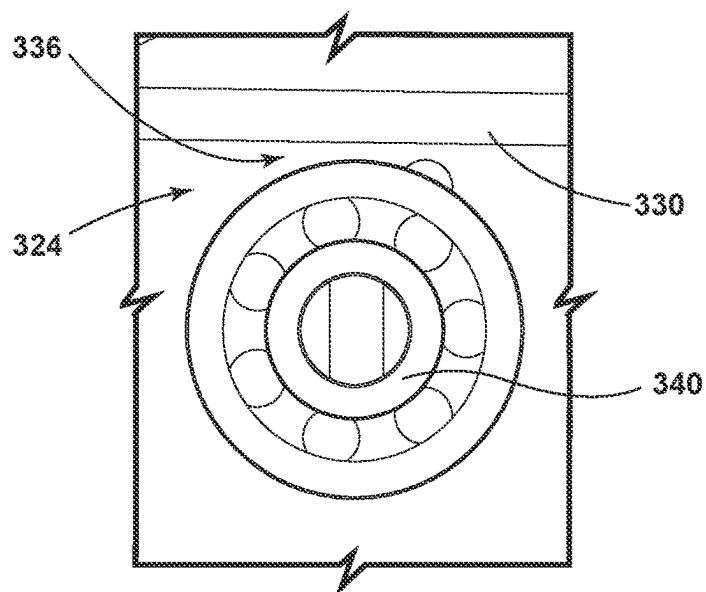


FIG. 15C

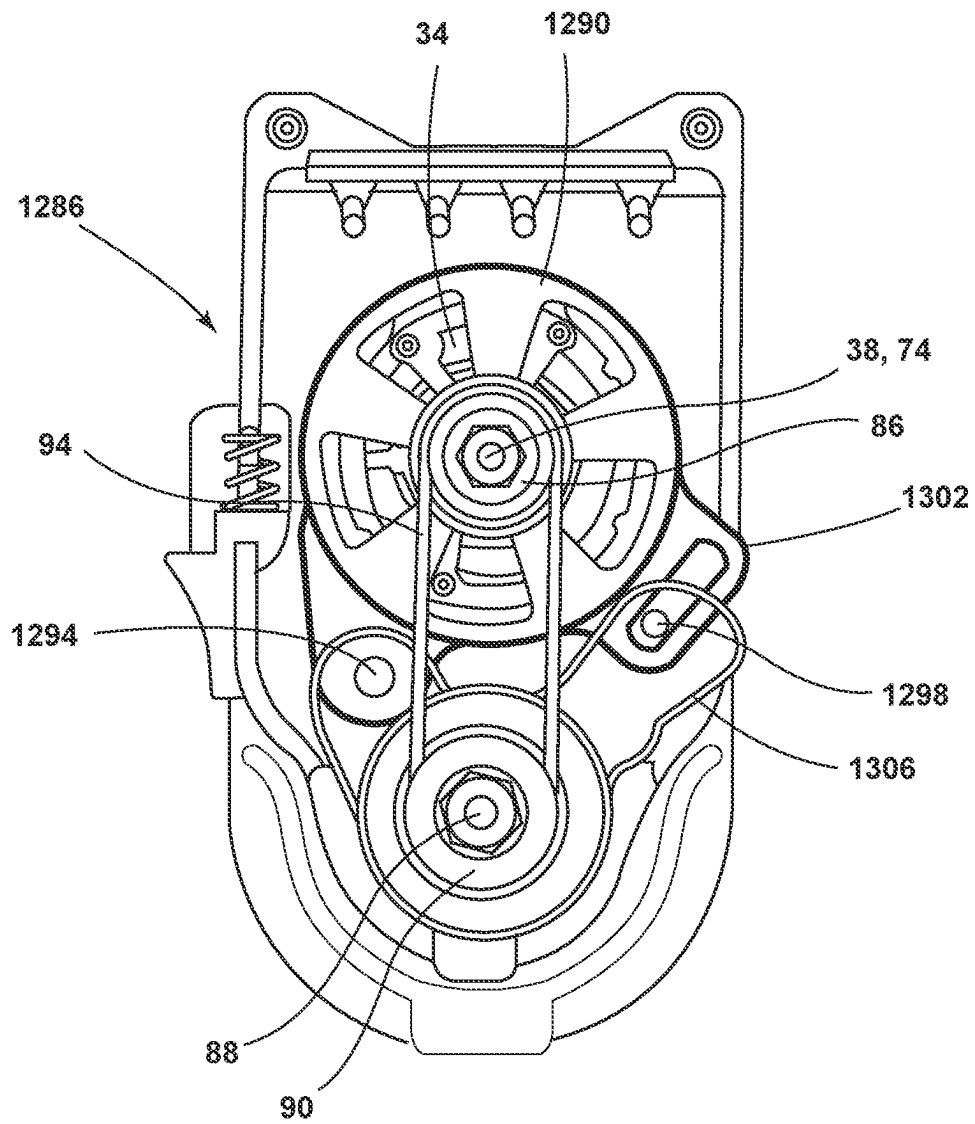
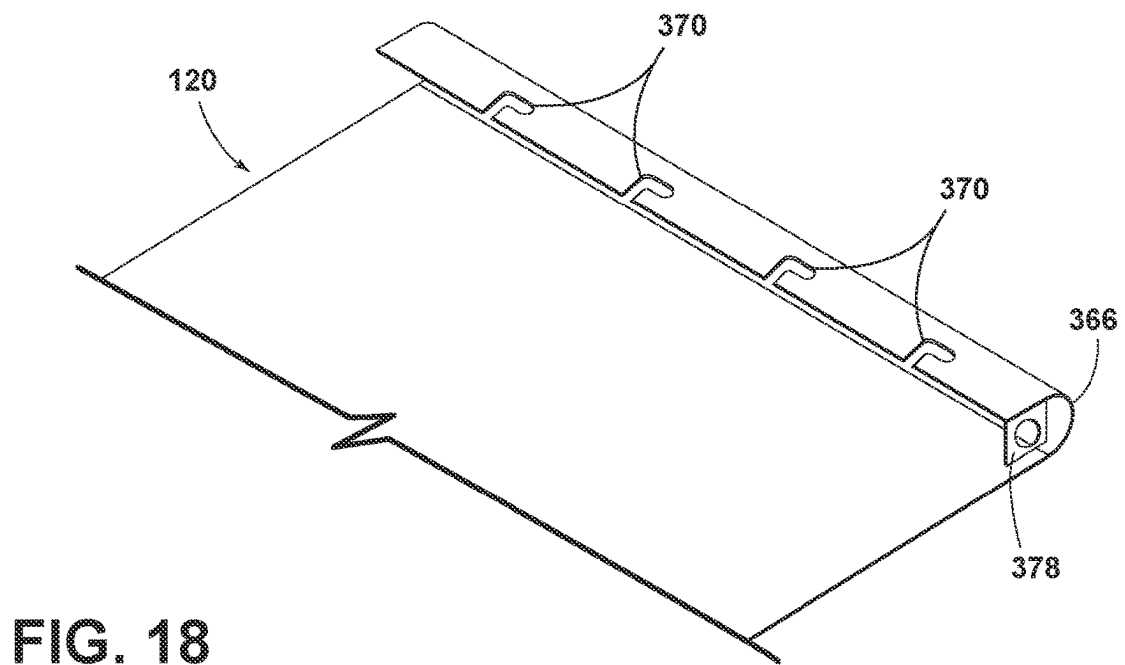
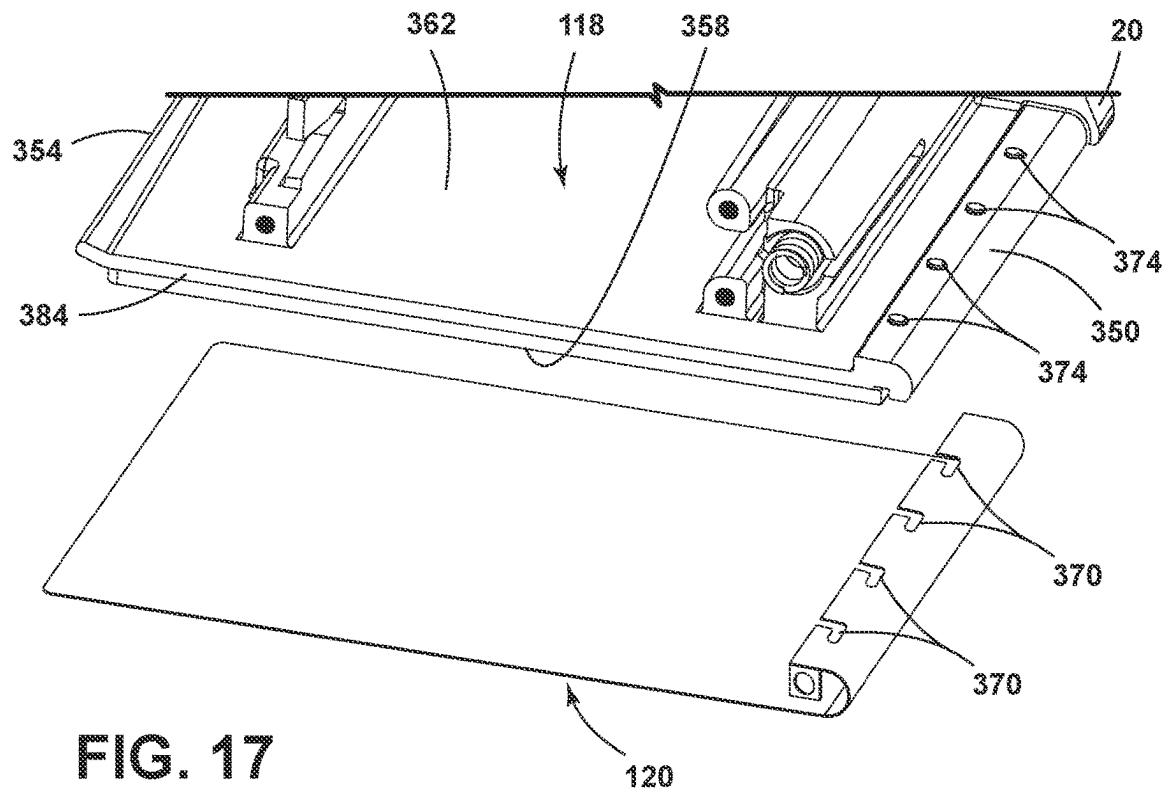
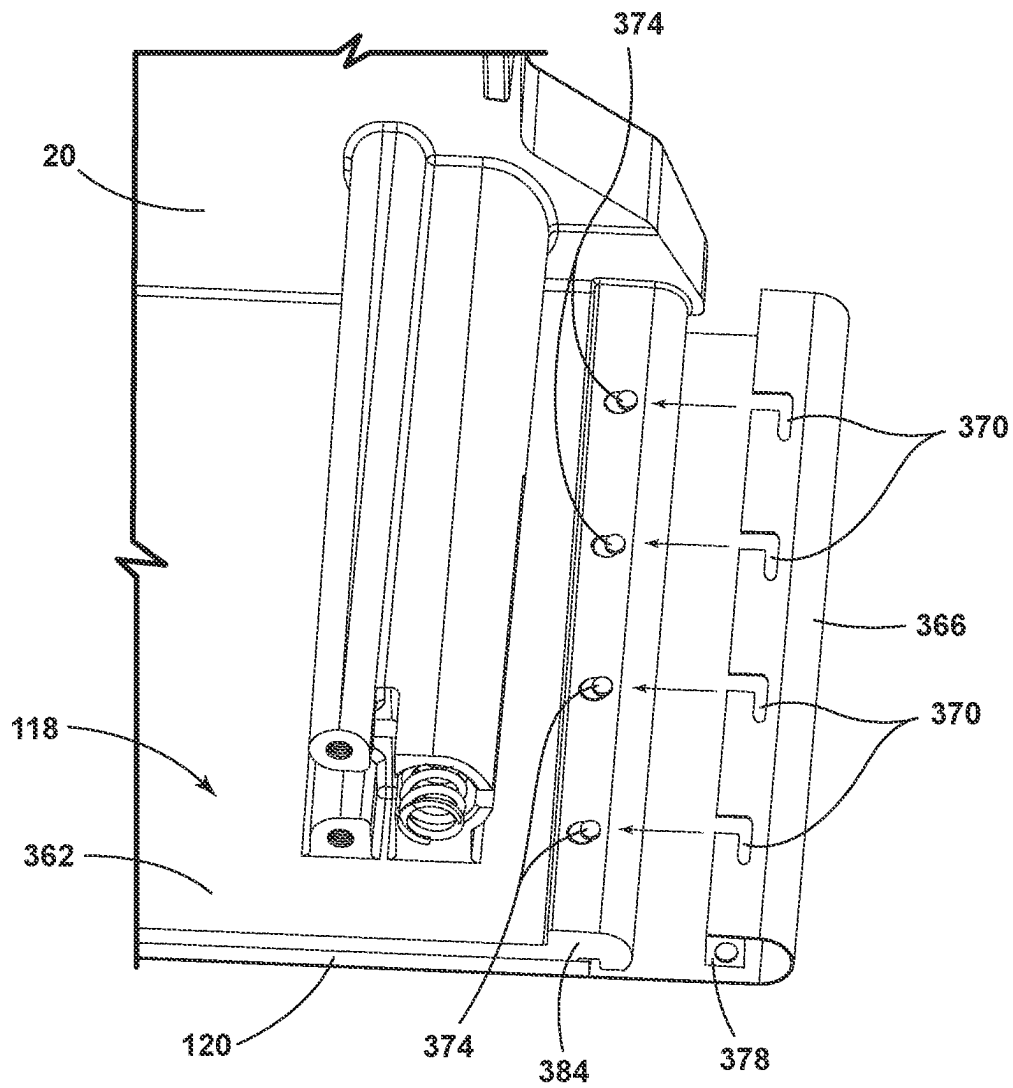


FIG. 16



**FIG. 19**

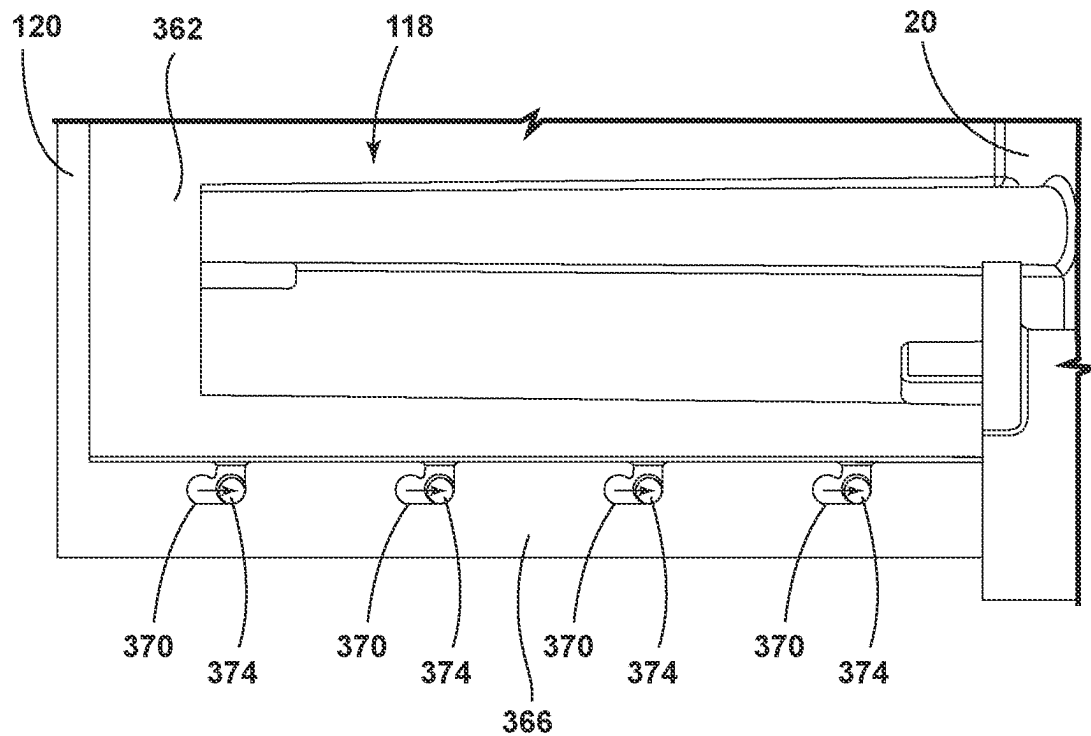


FIG. 20

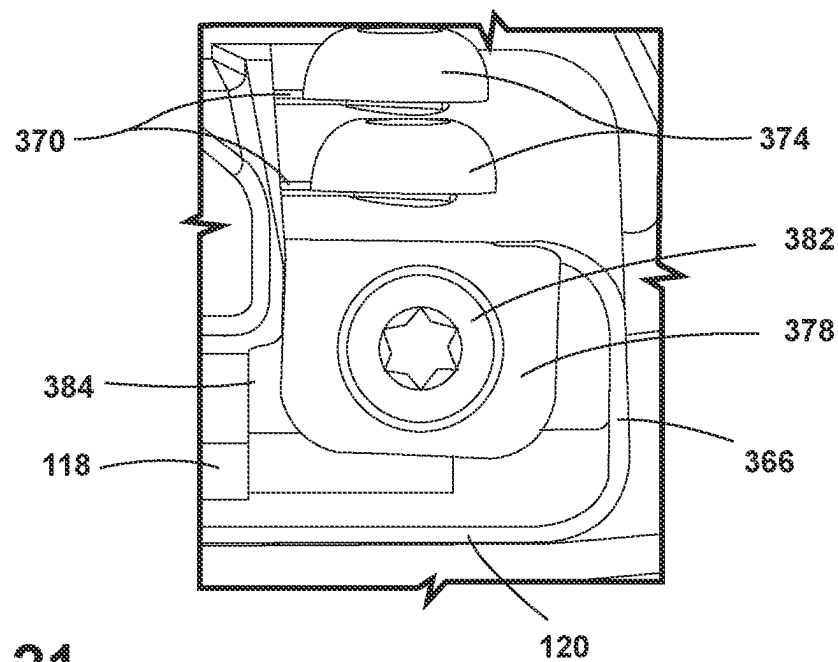
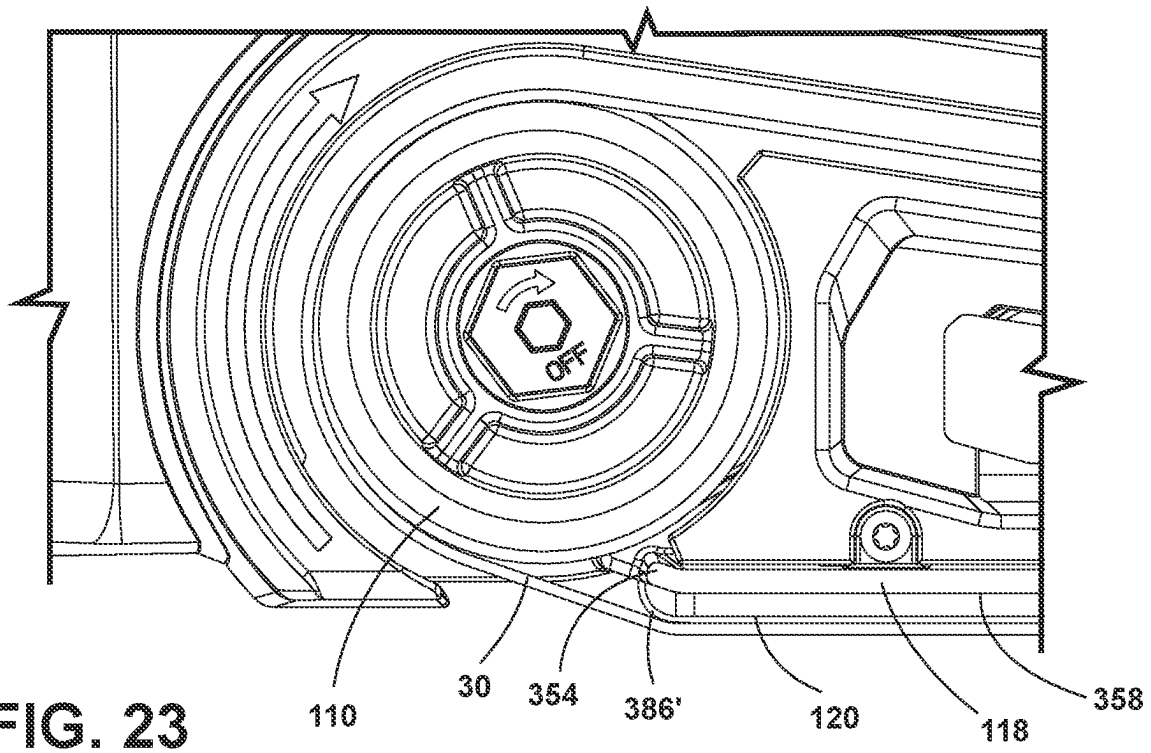
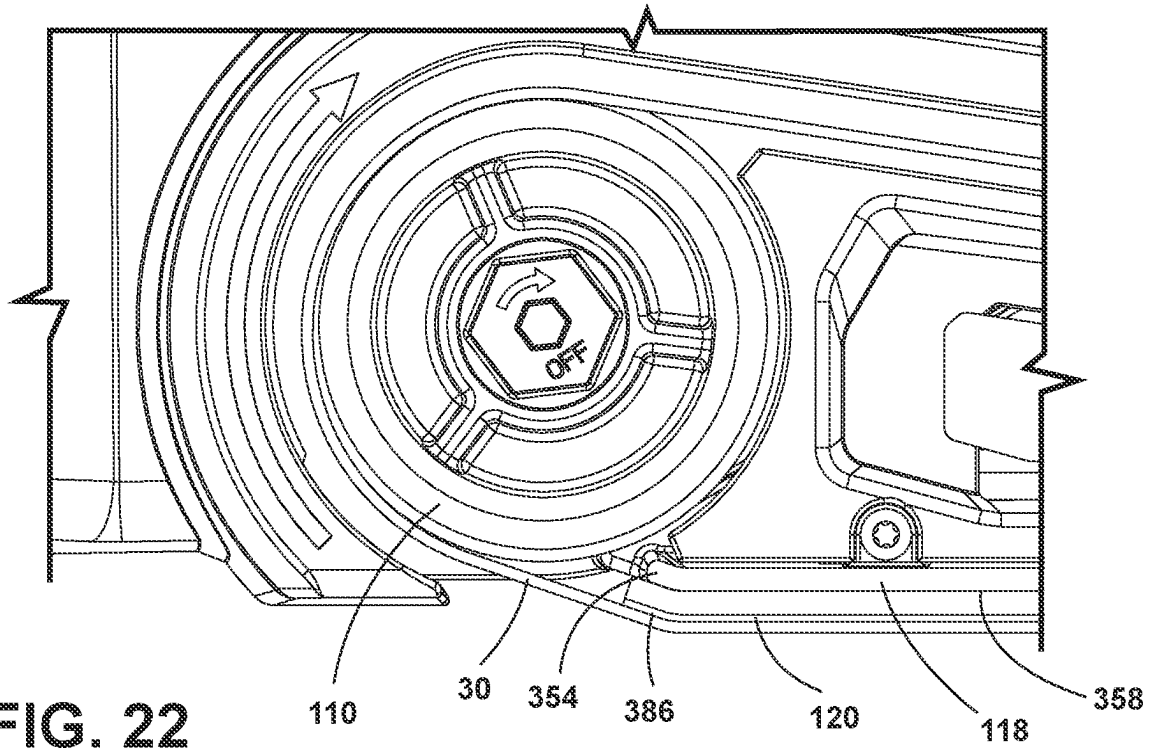
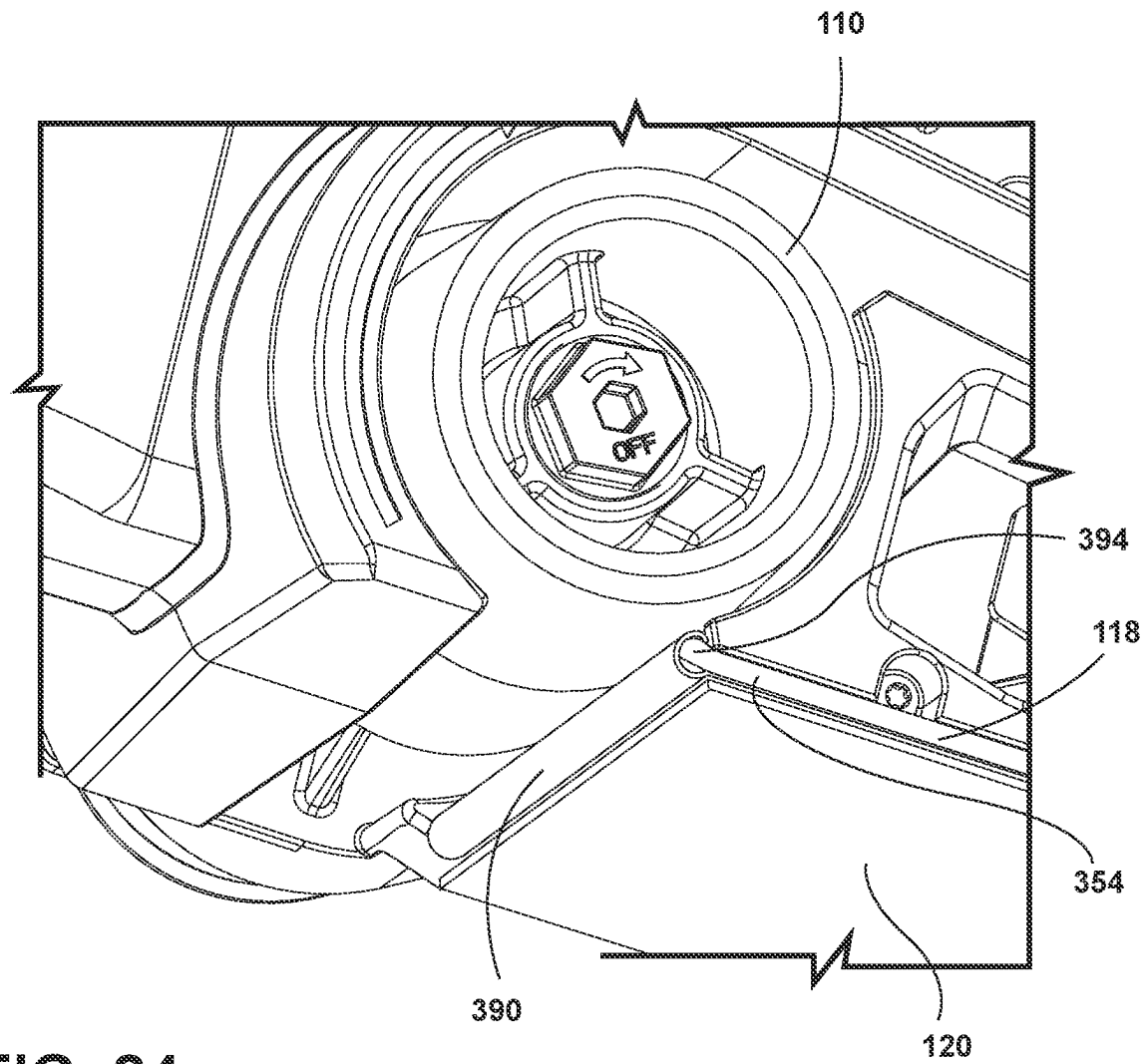


FIG. 21



**FIG. 24**

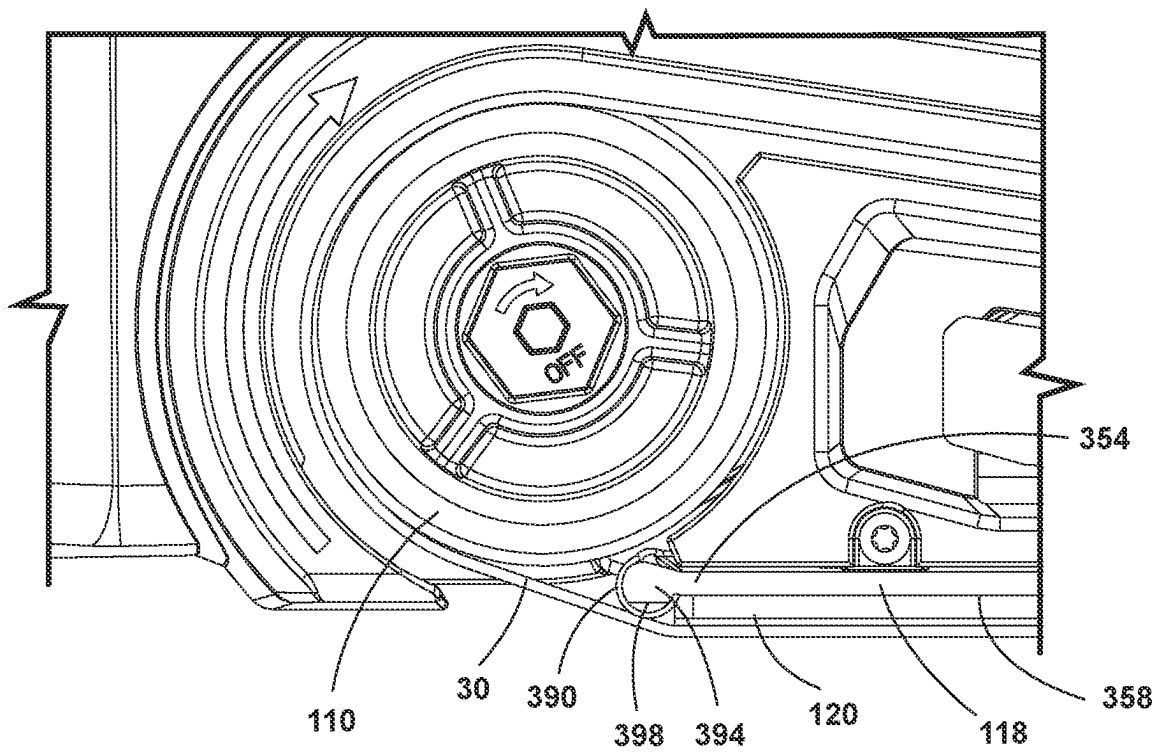


FIG. 25

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BELT SANDER**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims priority to U.S. Provisional Patent Application No. 63/301,212, filed Jan. 20, 2022, U.S. Provisional Patent Application No. 63/274,789, filed Nov. 2, 2021, U.S. Provisional Patent Application No. 63/246,655, filed Sep. 21, 2021, and U.S. Provisional Patent Application No. 63/173,545, filed Apr. 12, 2021, the entire contents of each of which are incorporated by reference herein.

FIELD OF THE INVENTION

The present invention relates to power tools, and in particular to portable belt sanders.

BACKGROUND OF THE INVENTION

Belt sanders generally include an abrasive sanding belt that is driven in a continuous loop. Typically, there is a series of drums that drive the sanding belt in a continuous loop, where the drums are spaced apart to create lateral runs therebetween. One of the lateral runs can be pressed against a workpiece to perform a sanding operation.

SUMMARY OF THE INVENTION

In one embodiment, the invention provides, among other things, a belt sander for sanding a workpiece and defining a center of gravity within a first plane. The belt sander includes a main housing, a handle defining a second plane, and a drive unit disposed within the main housing. The drive unit includes a motor disposed adjacent a first lateral sidewall of the main housing, a pulley system disposed adjacent a second lateral sidewall of the main housing opposite the first lateral sidewall, and a transmission disposed between the motor and the pulley system. The belt sander further includes a belt drive system driven by the drive unit. The belt drive system includes a drive wheel that is driven by the pulley system, a driven wheel that is driven by the drive wheel via a sanding belt, and a platen disposed between the drive wheel and the driven wheel to press the sanding belt against a workpiece. The platen defines a third plane extending perpendicular therefrom. The transmission and pulley system are disposed on one side of the first plane and the motor is disposed on an opposite side of the first plane. The motor is intersected by the second plane and the third plane.

In another embodiment, the invention provides, among other things, a belt sander for sanding a workpiece including a handle, and a main housing having a first lateral sidewall, a second lateral sidewall opposite the first lateral sidewall, and a front end being perpendicular to the first lateral sidewall. The belt sander further includes a drive unit disposed within the main housing between the first lateral sidewall and the second lateral sidewall, and a belt drive system driven by the drive unit. The belt drive system has a drive wheel, a driven wheel, and a sanding belt for engaging the workpiece. The belt sander further includes a first light disposed on the first lateral sidewall and a first lens for projecting light onto a first area of the workpiece that runs alongside the first lateral sidewall. The belt sander further includes a second light disposed on the front end and a second lens for projecting light onto a second area of the workpiece that is adjacent the driven wheel.

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In yet another embodiment, the invention provides, among other things, a belt sander for sanding a workpiece including a main housing, a handle extending from the main housing, a drive unit disposed within the main housing, a battery for selectively supplying electrical power to the drive unit, and a belt drive system driven by the drive unit. The belt drive system has a drive wheel, a driven wheel driven by the drive wheel via a sanding belt, and a platen for pressing the sanding belt against a workpiece while the sanding belt is rotated about the drive wheel and the driven wheel, creating dust and debris from the workpiece. The belt sander further includes a dust extraction unit driven by the motor transporting dust and debris away from the workpiece, and a dust bag positioned downstream of the dust extraction unit to receive dust and debris therefrom. The dust bag has a rectangular shape, and wherein the dust bag includes a zipper extending along first and second edges of the dust bag.

In still another embodiment, the invention provides, among other things, a belt sander for sanding a workpiece including a handle, a main housing having a first clamshell half and a second clamshell half that are secured together along a joint to form the main housing, a drive unit disposed within the main housing, and a battery for selectively supplying electrical power to the drive unit. The belt sander further includes a belt drive system driven by the drive unit. The belt drive system has a drive wheel, a driven wheel, and a sanding belt for engaging the workpiece. The belt sander further includes a first wire disposed within the first clamshell half and a second wire disposed within the second clamshell half. The first and second wires conduct electrical current from the battery to components within the belt sander. The belt sander further includes a wiring bridge that spans the joint between the first and second clamshell halves to electrically connect the first wire and the second wire. The wiring bridge is compressed when the first and second clamshell halves are coupled together.

In still yet another embodiment, the invention provides, among other things, a belt sander for sanding a workpiece including a main housing, a handle extending from the main housing, a drive unit disposed within the main housing, a battery for selectively supplying electrical power to the drive unit, and a first pulley system driven by the drive unit and having a first belt tensioner that is biased against a first pulley belt to remove excess slack from the first pulley belt. The belt sander further includes a belt drive system driven by the first pulley system and having a drive wheel, a driven wheel driven by the drive wheel via a sanding belt, and a platen for pressing the sanding belt against a workpiece while the sanding belt is rotated about the drive wheel and the driven wheel, creating dust and debris from the workpiece. The belt sander further includes a second pulley system driven by the drive unit and having a second belt tensioner that is biased against a second pulley belt to remove excess slack from the second pulley belt. The belt sander further includes a dust extraction unit driven by the second pulley system for transporting dust and debris away from the workpiece.

In still yet another embodiment, the invention provides, among other things, a belt sander for sanding a workpiece including a main housing, a handle extending from the main housing, a drive unit disposed within the main housing and including a motor that drives a drive shaft about a drive axis, and a battery for selectively supplying electrical power to the drive unit. The belt sander further includes a pulley system having a first pulley coupled to and driven by the drive shaft of the motor and a second pulley driven by the

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first pulley via a pulley belt. The belt sander further includes a belt drive system driven by the pulley system and having a drive wheel, a driven wheel driven by the drive wheel via a sanding belt, and a platen for pressing the sanding belt against a workpiece while the sanding belt is rotated about the drive wheel and the driven wheel. The belt sander further includes a belt tensioner for removing excess slack from the pulley belt, wherein the belt tensioner is capable of moving the first pulley relative to the second pulley during an adjustment state and inhibiting the first pulley from moving relative to the second pulley when the pulley belt is sufficiently tensioned during a locked state.

In still yet another embodiment, the invention provides, among other things, a belt sander for sanding a workpiece including a main housing, a handle extending from the main housing, a drive unit disposed within the main housing, a battery for selectively supplying electrical power to the drive unit, and a belt drive system driven by the drive unit. The belt drive system has a drive wheel, a driven wheel driven by the drive wheel via a sanding belt, and a platen defining a front edge disposed adjacent the driven wheel, a rear edge disposed adjacent the drive wheel, a bottom side disposed adjacent the workpiece, and a top side opposite the bottom side. The belt sander further includes a platen attachment that is removably coupled to the platen and is configured to press the sanding belt against a workpiece while the sanding belt is rotated about the drive wheel and the driven wheel. The platen attachment includes a lip that bends around at least one of the front edge or the rear edge from the bottom side to the top side, wherein the lip is coupled to the top side of the platen.

In still yet another embodiment, the invention provides, among other things, a belt sander for sanding a workpiece including a main housing, a handle extending from the main housing, a drive unit disposed within the main housing, a battery for selectively supplying electrical power to the drive unit, and a belt drive system driven by the drive unit. The belt drive system has a drive wheel, a driven wheel driven by the drive wheel via a sanding belt, and a platen defining a front edge disposed adjacent the driven wheel, a rear edge disposed adjacent the drive wheel, a bottom side disposed adjacent the workpiece, and a top side opposite the bottom side. The belt sander further includes a platen attachment that is removably coupled to the platen and is configured to press the sanding belt against a workpiece while the sanding belt is rotated about the drive wheel and the driven wheel. The belt sander further includes a wear skid that is removably coupled to the rear edge of the platen and protrudes beyond the platen attachment. The wear skid extends into the path of the sanding belt between the drive wheel and the platen attachment.

In still yet another embodiment, the invention provides, among other things, a belt sander for sanding a workpiece including a main housing, a handle extending from the main housing, a drive unit disposed within the main housing, a battery for selectively supplying electrical power to the drive unit, and a pulley system driven by the drive unit. The pulley system includes a first pulley, a second pulley, and a pulley belt having an upper run and a lower run. The belt sander further includes a vibration dampening system disposed adjacent to and selectively engaging the pulley system to inhibit oscillations within the pulley belt. The vibration dampening system includes at least one wave disruptor positioned away from outer periphery of the pulley belt, such that the pulley belt contacts the at least one wave disruptor when the upper run or the lower run of the pulley belt deviates from a straight-line path.

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Other aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a belt sander according to an embodiment of the invention.

FIG. 2 is a cross-sectional view of a drive unit within a main housing of the belt sander, taken along line 2-2 of FIG. 1.

FIG. 3 is another perspective view of the belt sander, illustrating a portion of the drive unit of FIG. 2.

FIG. 4 is a bottom perspective view of the belt sander of FIG. 1, illustrating a sanding belt disposed around a drive wheel, a driven wheel, and a platen.

FIG. 5 is a top plan view of the belt sander of FIG. 1, illustrating areas that are illuminated by work lights.

FIG. 6 is a cross-sectional view of a pommel, taken along line 6-6 of FIG. 5.

FIG. 7 is a side plan view of the belt sander of FIG. 1, illustrating the pommel in an unlocked position and slidable along a rail of the main housing.

FIG. 8 is a cross-sectional view of the pommel in a locked position, taken along line 8-8 of FIG. 5.

FIG. 9 is a plan view of a dust bag attachment mechanism for coupling a dust bag to the belt sander of FIG. 1.

FIG. 10 is a plan view of the dust bag for the belt sander, illustrating a zipper extending along edges of the dust bag.

FIG. 11 is a schematic of the zipper of the dust bag, illustrating a flap disposed behind the zipper on the interior of the dust bag.

FIG. 12A is a cross-section view of the main housing taken along line 6-6 of FIG. 5, illustrating a wiring bridge.

FIG. 12B is an enlarged perspective view of FIG. 12A, illustrating the wiring bridge.

FIG. 13 is a cross-sectional view of the wiring bridge taking along line 13-13 of FIG. 12B.

FIG. 14 is a perspective view of the belt sander, illustrating a first pulley system for driving the belt drive system.

FIG. 15A is another perspective view of the belt sander, illustrating a second pulley system for driving the dust extraction unit.

FIG. 15B is a plan view illustrating a vibration dampening system for the second pulley.

FIG. 15C is an enlarged plan view of a vibration dampening system in accordance with another embodiment, illustrating a bearing for the dampening system.

FIG. 16 is a plan view of the belt sander, illustrating the first pulley system for driving the belt drive system in accordance with another embodiment.

FIG. 17 is an exploded perspective view of the platen and a platen attachment configured to be coupled to the platen.

FIG. 18 is an enlarged perspective view of the platen attachment of FIG. 17, illustrating a plurality of slots disposed on a first lip of the platen attachment.

FIG. 19 is a perspective view of the platen attachment of FIG. 17 during assembly to the platen, illustrating the plurality of slots aligned with a plurality of pins of the platen.

FIG. 20 is a top plan view of the platen attachment during assembly to the platen, illustrating the plurality of pins received within the plurality of slots.

FIG. 21 is a side perspective view of the platen attachment, illustrating the platen attachment coupled to the platen via a fastener extending through a portion of the platen attachment.

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FIG. 22 is a side plan view of a platen attachment in accordance with another embodiment, illustrating the platen attachment coupled to the platen and including a second lip.

FIG. 23 is a side plan view of a platen attachment in accordance with another embodiment, illustrating the platen attachment coupled to the platen and including a second lip.

FIG. 24 is a rear perspective view of belt sander of FIG. 1, illustrating a wear skid removably coupled to the platen.

FIG. 25 is a side plan view of the belt sander of FIG. 1, illustrating the wear skid coupled to an enlarged head of the platen.

Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways.

DETAILED DESCRIPTION

FIG. 1 illustrates a portable power tool, such as a belt sander 10, for sanding a workpiece. In the illustrated embodiment, the belt sander 10 includes a main housing 14, a main frame 20 (FIG. 3) that supports the main housing 14, a primary handle 18 used for gripping and maneuvering the sander 10 along a workpiece, and a pommel or secondary handle 22 that is selectively grasped by a user to further stabilize the sander 10 during operation. The main housing 14 is comprised of two clamshell halves 16a, 16b that are connected together with threaded fasteners (e.g., screws), but may alternatively be secured together using other suitable coupling means.

With reference to FIGS. 1 and 2, the belt sander 10 further includes a drive unit 26 that is positioned within the main housing 14 and operable to drive a sanding belt 30 (FIG. 4). Specifically, the drive unit 26 includes an electric motor 34 (e.g., a brushless DC electric motor) capable of producing a rotational output through a drive shaft 38 which, in turn, provides a rotational input to a transmission 42. The motor 34 defines a motor axis 46 along which the drive shaft 38 and the transmission 42 are coaxially aligned. The belt sander 10 is powered by a battery pack 50, as shown in FIG. 5. The battery pack 50 connects to the primary handle 18 and provides electrical power to the motor 34 when a trigger 54 is depressed. The trigger 54 is conveniently positioned adjacent the primary handle 18 to allow a user to maneuver and activate the belt sander 10 with a single hand.

With reference to FIG. 2, the transmission 42 includes a transmission housing 58 affixed to the main frame 20, a single planetary gear stage 66 including a ring gear 62 positioned within the transmission housing 58 and three planetary gears 70 intermeshed with the ring gear 62. An output shaft 74 is coupled for co-rotation with a carrier 78 in the planetary gear stage 66 to thereby receive the torque output of the transmission 42. The output shaft 74 is coaxial with the motor axis 46. In other embodiments, the ring gear 62 receives the torque output of the transmission 42 to drive the output shaft 74. Although the transmission 42 of the illustrated embodiment includes a single planetary gear stage 66, in other embodiments, the transmission 42 may alternatively include two or more planetary gear stages. Still in other embodiments, the transmission 42 may alternatively include one or more spur gear sets, one or more helical gear sets, or other styles of transmissions.

With reference to FIGS. 2 and 3, the drive unit 26 further includes a first pulley system 82 that is driven by the output

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shaft 74 of the transmission 42. The first pulley system 82 is downstream of the transmission 42, such that the transmission 42 is positioned between the motor 34 and the first pulley system 82 in a direction along the motor axis 46. As a result, the output shaft 74 (and therefore, the first pulley system 82) has a reduced angular velocity compared to the drive shaft 38 due to the gear reduction of the transmission 42. The first pulley system 82 includes a first pulley 86 coupled for co-rotation with the output shaft 74 and a second pulley 90 driven by the first pulley 86 via a pulley belt 94 (FIG. 3) about a driven shaft 88. Both the first pulley 86 and the second pulley 90 are supported by the main frame 20. On the outer periphery of the first and second pulley 86, 90 are teeth 98 that intermesh with corresponding teeth 102 of the pulley belt 94 to provide a positive engagement therebetween and a synchronous drive between the belt 94 and the pulleys 86, 90. The second pulley 90 is coupled to a drive belt system 106 (FIG. 4) that drives the sanding belt 30, as described in further detail below.

With reference to FIGS. 2-4, the drive belt system 106 includes a drive wheel 110, a driven wheel 114 driven by the drive wheel 110 via the sanding belt 30, and a platen 118 disposed between the drive wheel 110 and the driven wheel 114. In some embodiments, the platen 118 is integrally formed with the main frame 20 (FIG. 15A). When the sanding belt 30 is being driven by the drive wheel 110, the sanding belt 30 slides along the platen 118 (or a platen attachment 120 coupled to the platen 118, as shown in FIG. 17), creating a flat, working surface 122 that presses the sanding belt 30 against a workpiece. The drive wheel 110 is disposed adjacent a rear end 124 of the belt sander 10 and defines a drive wheel axis 126, whereas the driven wheel 114 is disposed adjacent a front end 128 of the belt sander 10 and defines a driven wheel axis 130. As illustrated in FIG. 2, the motor axis 46, the drive wheel axis 126, and the driven wheel axis 130 are all parallel. A tensioning mechanism 134 is coupled to the driven wheel 114 and is capable of moving the driven wheel 114 between a retracted position, in which the sanding belt 30 may be removed from the sander 10, and an extended position, in which the sanding belt 30 is placed under tension and inhibited from being removed from the sander 10. Specifically, the driven wheel 114 moves toward the drive wheel 110 in the retracted position to allow the sanding belt 30—having a fixed, rigid circumference—to fit around the drive wheel 110 the driven wheel 114, and the platen 118.

Returning to FIG. 2, the components of the belt sander 10 are situated in a specific arrangement to improve the center of gravity, ergonomics, longevity, and performance of the belt sander 10, as described in further detail below. The main housing 14 includes a first lateral sidewall 138 (as represented by broken line in FIG. 2) and a second lateral sidewall 142 opposite the first lateral sidewall 138. The belt sander 10 defines a center of gravity that is located between first and second lateral sidewalls 138, 142. Specifically, the center of gravity for the belt sander 10 is located within a first upright plane 146 that is approximately equal distance from the first and second lateral sidewalls 138, 142. By positioning the transmission 42 between the motor 34 and the first pulley system 82, the plane 146 (which contains the center of gravity) is situated centrally within the belt sander 10 because the weight of the motor 34 is positioned proximate the first lateral sidewall 138, thereby counteracting the weight of the first pulley system 82 positioned proximate the second lateral sidewall 142 without increasing the length of the drive shaft 38. That is, the transmission 42 enables the motor 34 and the first pulley system 82 to be positioned

further apart (i.e., near opposite lateral sidewalls **138**, **142**) without compromising the integrity of the drive shaft **38**. As a result, the transmission **42** avoids unnecessary length extensions of the drive shaft **38**, which can inadvertently cause excessive shaft vibration, reduced tool balance, excessive torsional deflection, reduced shaft longevity, and reduced longevity of connected components (e.g., the motor **34**, the first pulley system **82**, etc.).

With continued reference to FIG. 2, the primary handle **18** defines a second upright plane **150** that extends through the center of the primary handle **18**, while the platen **118** defines a third upright plane **154** that extends through the center of the platen **118** in a direction perpendicular to the platen **118**. Each plane **146**, **150**, **154** is parallel and substantially perpendicular relative to a support surface (e.g., the ground). As shown in FIG. 2, the first upright plane **146** is closer to the second upright plane **150** than the third upright plane **154**. Also, the first lateral sidewall **138** is closer to the third upright plane **154** than the second upright plane **150**. In this configuration, the motor **34** is intersected by both the second upright plane **150** and the third upright plane **154**. By positioning the motor **34** in this manner, the motor **34** is adjacent the first lateral sidewall **138** while the first pulley system **82** is adjacent the second lateral sidewall **142**, thereby providing sufficient volume within the main housing **14** to accommodate the transmission **42**. This configuration of components (e.g., the motor **34**, the transmission **42**, and the first pulley system **82**) also moves the first upright plane **146**, and therefore the center of gravity, towards the third upright plane **154**, which optimizes weight distribution on the working surface **122**. As a result, the reliability of the belt sander **10** is increased, and the vibration and noise emitted by the drive unit **26** is also decreased. Further, the ergonomics of the belt sander **10** is improved due to the optimized center of gravity.

With reference to FIGS. 1 and 5, the belt sander **10** further includes a first light **158** with a first lens **162** disposed on the first lateral sidewall **138** and a second light **166** with a second lens **170** disposed on the front end **128**. The lights **158**, **166** provide a user increased visibility during operation, especially in instances where light may be limited (e.g., dark corners of a workpiece). In the illustrated embodiment, the lights **158**, **166** are light-emitting diodes (LEDs), but may alternatively be different types of lights in other embodiments.

Referring to FIG. 5, the first lens **162** is a convex lens, such that light emitted from the LED diverges outward from the first lateral sidewall **138** and projects downward onto area **A1** of a workpiece that runs alongside the first lateral sidewall **138**. Similarly, the second lens **170** is a convex lens, such that light emitted from the LED diverges outward from the front end **128** and projects downward onto area **A2** of a workpiece that is adjacent the driven wheel **114**. The second lens **170** also projects light outward in front of the belt sander **10** onto area **A3** of a workpiece to reveal deformities and finish quality on a workpiece. Specifically, the second light **166** is positioned in the lower half of the main housing **14** on the front end **128**, such that the incident ray (i.e., light rays coming from the second light **166**) has an acute angle of incidence relative to a workpiece, creating shadows on a workpiece where deformities may exist in front of the front end **128**. Generally, the absence of shadows on a workpiece indicates absence of deformities (e.g., bumps, divots, scratches, etc.). Lastly, the first and second lights **158**, **166** assist a user from inadvertently running into and damaging (e.g., marring) walls of a workpiece due to lack of light.

With reference to FIGS. 6-8, the secondary handle **22** of the belt sander **10** is moveable relative to the main housing **14**. Specifically, the main housing **14** includes a rail **174**, upon which the secondary handle **22** is moveably mounted along a rail axis **178** (FIG. 7). The secondary handle **22** includes a lock system **180** that is pivotable between a locked state, in which the secondary handle **22** is inhibited from moving along the rail **174**, and an unlocked state, in which the secondary handle **22** is permitted to move along the rail **174**.

With continued reference to FIGS. 6-8, the lock system **180** includes an axle **184** (FIG. 8) received within corresponding counterbores **186** in the secondary handle **22**, a cam body **188** disposed around the axle **184** (FIG. 6), and a lever **192** extending away from the cam body **188** and capable of rotating the cam body **188**. In the locked state, the lever **192** is substantially flush (i.e., flush, slightly sub-flush, or slightly proud, etc.) with the geometry of the secondary handle **22**, such that a user does not feel the lever **192** when it is in the locked state. However, at a distal end of the lever **192** is a lip **194** that extends beyond the secondary handle **22** to allow a user to grasp the lever **192** when the lock system **180** is in the locked state. Furthermore, the cam body **188** (FIG. 6) is received within one of two arcuate recesses **196a**, **196b** on the rail **174** when the lock system **180** is in the locked state. As such, the secondary handle **22** is moveable along the rail **174** between two discrete positions—a rearward position (as represented by phantom lines in FIG. 7) and a forward position (as represented by solid lines in FIG. 7). In the rearward position, the cam body **188** is received within the arcuate recess **196a**. In the forward position, the cam body **188** is received within arcuate recess **196b**.

Not only does the cam body **188** mechanically interfere with the rail **174** to inhibit movement of the secondary handle **22** relative to the rail **174**, but the cam body **188** also mechanically interferes with the arcuate recesses **196a**, **196b** when received therein. The mechanical interference between the cam body **188** and the arcuate recesses **196a**, **196b** causes the lock system **180** to displace upward relative to the rail **174** against the bias of a spring **198** (FIG. 8). As a result of the lock system **180** displacing upwardly, the spring **198** deforms and a force component **F1** is exerted on a track **200** of the secondary handle **22**, causing the secondary handle **22** to displace in a direction substantially perpendicular to the rail axis **178**. Subsequently, the slop (i.e., play) or relative movement between the track **200** of the secondary handle **22** and the rail **174** is limited when the lock system **180** is in the locked state. At this point, there is increased surface area contact between the track **200** and the rail **174**, which increases friction and inhibits the secondary handle **22** from moving. In the illustrated embodiment, the spring **198** is an O-ring disposed between the axle **184** and each counterbore **186** of the secondary handle **22**. The spring **198** (or O-ring) is composed of an elastomeric material, allowing for relative movement between the axle **184** and the counterbore **186** of the secondary handle **22**. In other embodiments, the spring **198** may alternatively be a compression spring, an extension spring, or other elastic material capable of elastic deformation.

With reference to FIG. 7, the lock system **180** further includes a plateau **202**, which is a flat region disposed on the cam body **188** that creates a discontinuity on the cam body **188**. The plateau **202** removes the cam body **188** from the rail **174** when the lock system **180** is in the unlocked state. Specifically, the plateau **202** is disposed on the cam body **188**, such that the cam body **188** is spaced away from the rail **174** when the lock system **180** is in the unlocked state. In

other words, the cam body **188** is removed from one of the arcuate recesses **196a**, **196b** and the plateau **202** is substantially parallel to the rail axis **178** when the lock system **180** is actuated to the unlocked state. At this point, the track **200** of the secondary handle **22** is capable of sliding along the rail **174** between the rearward position and the forward position. Once the secondary handle **22** is moved to either the rearward position or the forward position, the plateau **202** is aligned with the respective arcuate recess **196a**, **196b**, at which point the lock system **180** may be actuated to the locked state where the cam body **188** rotates into the respective arcuate recess **196a**, **196b**. In the unlocked state, the lever **192** experiences rotational resistance due to frictional forces as the axle **184** slides within the spring **198** (i.e., O-ring).

With reference to FIG. 3, the belt sander **10** further includes a dust extraction unit **206** coupled to and extending from the main housing **14**. The dust extraction unit **206** draws dust particles and other debris away from a workpiece and collects the debris in a dust bag **210** during operation. The dust extraction unit **206** includes a shroud **214** and a fan **218** disposed within the shroud **214**. The shroud **214** includes an inlet **220** that is adjacent the platen **118** and an outlet **222** that is adjacent the dust bag **210**. The fan **218** is driven by the motor **34** via a pulley belt **224** of a second pulley system **226** (including a first pulley **228** and second pulley **229**; FIG. 15A), such that the fan **218** is configured to create a low-pressure region near the platen **118** to draw debris into the inlet **220**. As shown in FIGS. 2 and 3, the second pulley system **226** is coupled to the drive shaft **38** upstream of the transmission **42** via the first pulley **228**. That said, the transmission **42** is disposed between the first pulley system **82** and the second pulley system **226**. As a result, the first pulley system **82** and the second pulley system **226** are coaxially driven along the motor axis **46**, albeit at different speeds. Specifically, the first pulley system **82** is driven at a first speed that is identical to the angular velocity of the output shaft **74**, whereas the second pulley system **226** is driven at a second speed that is identical to the angular velocity of the drive shaft **38**. The angular velocity of the second speed is greater than the angular velocity of the first speed due to the gear reduction of the transmission **42**. Although the first and second pulley systems **82**, **226** are being coaxially driven along the motor axis **46**, in other embodiments, the first and second pulley systems **82**, **226** may alternatively be driven along parallel axes.

With reference to FIGS. 14 and 15, the first pulley system **82** further includes a first belt tensioner **286** and the second pulley system **226** includes a second belt tensioner **290**. The first belt tensioner **286** is mounted to the main frame **20** and disposed within the inner periphery of the pulley belt **94** of the first pulley system **82**. The first belt tensioner **286** includes an axle **294** coupled to the main frame **20**, an arm **298** coupled to and extending perpendicularly relative to the axle **294**, and a first roller **302** rotatably coupled to the arm **298**. The first roller **302** extends away from the arm **298** into the path of the pulley belt **94**, such that the first roller **302** is in contact with the pulley belt **94**. A spring **306** (e.g., a torsion spring, etc.) is disposed around the axle **294** and biases the arm **298** in a clockwise direction (frame of reference from FIG. 14) to bias the first roller **302** into contact with the inner periphery of the pulley belt **94**.

With reference to FIG. 15A, the second belt tensioner **290** is also mounted to the main frame **20** and disposed within the inner periphery of the pulley belt **224** of the second pulley system **226**. The second belt tensioner **290** includes an axle **310** coupled to the main frame **20**, an arm **314**

coupled to and extending perpendicularly relative to the axle **310**, and a second roller **318** rotatably coupled to the arm **314**. The second roller **318** extends away from the arm **314** into the path of the pulley belt **224**, such that the second roller **318** is in contact with the pulley belt **224**. A spring **322** (e.g., a tension spring, etc.) is coupled between the main frame **20** and the arm **314**, and biases the arm **314** in a counterclockwise direction (frame of reference from FIG. 15A) to bias the second roller **318** into contact with the inner periphery of the pulley belt **224**. The first belt tensioner **286** and the second belt tensioner **290** apply tension to the respective pulley belts **94**, **224**, thereby reducing belt tension variation and prolonging the longevity of the pulley belts **94**, **224**.

In other embodiments, the belt sander **10** may alternatively include a belt tensioner **1286** that is configured to adjust the tension of at least one of the pulley belts **94**, **224**. The belt tensioner **1286** includes the drive shaft **38**, the output shaft **74**, the first pulley **86**, the driven shaft **88**, and the second pulley **90**. In some embodiments, the belt tensioner **1286** may also include at least one of the motor **34** and the transmission housing **58**. With reference to FIG. 16, the motor **34** drives the drive shaft **38** which, in turn, drives the output shaft **74** and the first pulley **86**. The first pulley **86** is coupled to and drives, for example, the pulley belt **94** and the second pulley **90** via the pulley belt **94**. The motor **34** is disposed within a motor housing **1290**, which is pivotably coupled within the main housing **14** about a pin **1294** such that the motor axis **46** is movable relative to the axis of the driven shaft **88**. Specifically, the motor axis **46** is capable of being moved towards or away from the axis of the driven shaft **88** along an arcuate path in response to the motor housing **1290** pivoting about the pin **1294**. If the motor axis **46** is moved towards the axis of the driven shaft **88**, then slack is introduced into the pulley belt **94**. If, on the other hand, the motor axis **46** is moved away from the axis of the driven shaft **88**, then slack is removed and the pulley belt **94** is tensioned. When the motor housing **1290** is pivoted about the pin **1294** to the desired location, the motor housing **1290** is fixed into position via a fastener **1298**. In the illustrated embodiment, the motor housing **1290** includes a motor bracket **1302** extending away from the motor housing **1290** that receives the fastener **1298**. In some embodiments, the motor housing **1290** is fixed into position by rigidly interconnecting the motor housing **1290** and a bracket **1306** of the second pulley **90** via the fastener **1298**. Although not illustrated, the belt tensioner **1286** may also be employed for tensioning the pulley belt **224**. Also not illustrated, the belt tensioner **1286** may include a handle or actuator disposed outside the main housing **14** so that a user can manipulate the handle or actuator to adjust the distance between the first pulley **86** and the second pulley **90**.

As illustrated in FIG. 15B, the belt sander **10** may also include a vibration dampening system **324**. In some embodiments, the vibration dampening system **324** may be in addition to the belt tensioner **286**, **290**, or replace the belt tensioner **286**, **290** altogether. The vibration dampening system **324** includes a plurality of wave disruptors **326**, **328** disposed adjacent the pulley belt **224** and coupled to the main frame **20**. The wave disruptors **326**, **328** are configured to cancel or disrupt any waves in the pulley belt **224** that form as a result of vibration in the belt sander **10**. In the illustrated embodiment of FIG. 15B, the first wave disruptor **326** is positioned adjacent the first pulley **228**, whereas the second wave disruptor **328** is positioned adjacent the second pulley **229**. Furthermore, the first wave disruptor **326** is disposed adjacent a lower run **330** of the pulley belt **224**,

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whereas the second wave disruptor **328** is disposed adjacent an upper run **332** of the pulley belt **224**. Additionally, each wave disruptor **326**, **328** is positioned away from the outer periphery of the pulley belt **224** to avoid contact with the pulley belt **224**. Specifically, the first and second wave disruptors **326**, **328** are positioned a small distance away from the lower run **330** and the upper run **332**, respectively, thereby defining a first air gap **336** between the first wave disruptor **326** and the lower run **330**, and defining a second air gap **338** between the second wave disruptor **328** and the upper run **332**. In other words, the first and second wave disruptors **326**, **328** are not in immediate contact with the pulley belt **224** when the lower and upper runs **330**, **332** are traveling along their respective straight-line paths **342**, **344**. The air gaps **336**, **338** are intended to avoid excessive rubbing and heat generation between the pulley belt **224** and the wave disruptors **326**, **328**, resulting in increased lifespan of the pulley belt **224**. Although the vibration dampening system **324** of the illustrated embodiment is disposed adjacent the pulley belt **224**, in other embodiments, the vibration dampening system **324** may also be incorporated adjacent the pulley belt **94**.

The wave disruptor **326** is only capable of engaging the lower run **330** at a single contact point. Likewise, the wave disruptor **328** is only capable of contacting the upper run **332** at a single contact point.

During operation, the lower run **330** and the upper run **332** travel along the respective straight-line paths **342**, **344** between the first pulley **228** and the second pulley **229**. Specifically, the lower run **330** travels from the first pulley **228** to the second pulley **229** along the straight-line path **342**, while the upper run **332** travels from the second pulley **229** to the first pulley **228** along the straight-line path **344**. In some instances, however, the pulley belt **224** may begin to oscillate or vibrate, such that the lower run **330** and the upper run **332** no longer travel along the straight-line paths **342**, **344**. Rather, the lower run **330** and the upper run **332** may begin to form waves (e.g., sinusoidal wave forms) between the first and second pulley **228**, **229**. These oscillations or waves of the pulley belt **224** can cause unwanted damage and wear to the pulley belt **224** and may reduce the lifespan of the pulley belt **224**. When a single oscillation (e.g., wave) of the lower run **330** reaches an amplitude (i.e., perpendicular height from the straight-line path **342** to the crest of the wave) equivalent to the first air gap **336**, the first wave disruptor **326** is capable of contacting the lower run **330**, thereby interrupting the wave so the lower run **330** can travel along the straight-line path **342** again. Additionally, when the oscillation (e.g., wave) of the upper run **332** reaches an amplitude (i.e., perpendicular height from straight-line path **344** to the crest of the wave) equivalent to the second air gap **338**, the second wave disruptor **328** contacts the upper run **332**, thereby interrupting the wave so the upper run **332** can travel along the straight-line path **344** again. Therefore, the vibration dampening system **324** is engageable with the pulley belt **224** to inhibit oscillations within the pulley belt **224** and maintain the pulley belt **224** traveling along the straight-line paths **342**, **344** to increase the lifespan of the pulley belt **224**.

In the illustrated embodiment of FIG. 15B, the first and second wave disruptors **326**, **328** are first and second pins, respectively. Each pin is merely a cylindrical pin that is press-fit into the frame **20**, such that the pin is a stationary component that does not rotate. The pins may alternatively be another type of mechanical component. For example, FIG. 15C illustrates that the wave disruptors **326**, **328** are first and second bearings **340**, respectively. Each bearing

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340 is disposed on a shaft, allowing the bearing **340** to rotate when the pulley belt **224** contacts the bearing **340**. The bearing **340** can reduce friction, and therefore, increase the lifespan of the pulley belt **224**. Still, in further embodiments, the wave disruptors **326**, **328** may alternatively be bushings, ceramic rollers, or other similar type of mechanical component.

With reference to FIG. 9, the dust bag **210** is removably coupled to the outlet **222** of the shroud **214**. Specifically, the dust bag **210** includes a conduit inlet **230** and a quick-disconnect mechanism **234** that connects with the outlet **222** to secure the dust bag **210** onto the belt sander **10**. The conduit inlet **230** fits over the outlet **222** (e.g., press fit, slip fit, etc.) while the quick-disconnect mechanism **234** is abutted with a shoulder **238** of the outlet **222**. The shoulder **238** is annularly disposed around and extends away from the outlet **222**, allowing the latch **242** to overlap with the shoulder **238** regardless of the orientation of the dust bag **210**.

The quick-disconnect mechanism **234** includes a latch **242** that is biased towards the shoulder **238** via a compression spring **246**. In other embodiments, the spring **246** is a torsional spring, a tension spring, or another type of spring. When the latch **242** overlaps the shoulder **238**, the dust bag **210** is inhibited from inadvertent removal from the belt sander **10**. A user, however, can pivot the latch **242** away from the shoulder **238** (as represented by broken lines of FIG. 9) by depressing a button **250** against the bias of the spring **246**, causing the latch **242** to be spaced away from the shoulder **238** and permitting removal of the dust bag **210** from the outlet **222**. In other embodiments, the latch **242** is an elastic body (e.g., a snap latch) that is capable of elastic deformation as the latch **242** slides over the shoulder **238** to permit attaching and removing the dust bag **210** to the belt sander **10**. Once the dust bag **210** is removed from the belt sander **10**, a user may empty the contents of the dust bag **210** into a waste receptacle, as described in further detail below.

With reference to FIG. 10, the dust bag **210** further includes a zipper **254** for allowing access into the dust bag **210** and removal of dust and debris. The dust bag **210** is approximately rectangular in shape with the zipper **254** extending along the full length of two perpendicular edges **258a**, **258b** of the dust bag **210**. The other two edges **258c**, **258d** are sewn closed with the conduit inlet **230** extending along and supporting the edge **258d**. The conduit inlet **230** also provides structure to the dust bag **210** so the dust bag **210** does not collapse and clog airflow during operation. With the zipper **254** extending along the full length of the edges **258a**, **258b**, dust and debris within the dust bag **210** can be quickly emptied and disposed in a waste receptacle, leaving behind minimal debris in the corners of the dust bag **210** without heavy shaking of the dust bag **210**. Although the zipper **254** of the illustrated embodiment is a single zipper that extends along edges both **258a**, **258b**, in other embodiments, multiple zippers may alternatively be used that each extend along the respective edges.

With reference to FIG. 11, the dust bag **210** further includes a flap **262** of extra fabric disposed along the zipper **254**. Specifically, the flap **262** is disposed on the inside of the dust bag **210** and shields the zipper **254** from dust particles and debris that can otherwise escape through the teeth of the zipper **254**, causing the zipper **254** to clog and jam. As such, the flap **262** extends at least along the entire length **L1** of the zipper **254**, as well as extends at least the entire width **W1** of the zipper **254**. In fact, a length **L2** of the flap **262** is longer than the length **L1** of the zipper and a width **W2** of the flap **262** is wider than the width **W1** of the zipper **254**.

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Thus, for debris to escape through the zipper **254**, the debris must first navigate around the flap **262** and through the teeth of the zipper **254**. The flap **262** inhibits dust particles from escaping the dust bag **210** and reduces the likelihood that the zipper **254** will clog from a buildup of dust particles between the teeth of the zipper **254**.

With reference to FIGS. **12A-13**, the belt sander **10** further includes a wiring bridge **266** that conducts electrical current across a joint **268** (FIG. **13**) between the clamshell halves **16a**, **16b**. To provide some background, the routing of wires in tools and other devices is often distributed throughout both clamshell halves. Oftentimes, some wire routing is required to span across the separate clamshell halves **16a**, **16b** in order to electrically connect all components within the tool or device. So, a wire is used to span across the joint between the two clamshell halves, resulting in blind wire routing during assembly and possible inadvertent damage to the wire or other components.

With continued reference to FIGS. **12A-13**, the clamshell half **16a** includes an electrical processor **270** with a first set of wires **274a**, **274b** extending therefrom to transport electrical current and signals to/from various components of the belt sander **10** (e.g., motor **34**, battery pack **50**, trigger **54**, first light **158**, etc.). The other clamshell half **16b** includes a second set of wires **278a**, **278b** that are capable of electrically communicating with the first set of wires **274a**, **274b** (FIG. **13**) and other components coupled to the clamshell half **16b** (e.g., second light **166**, etc.). The wiring bridge **266** includes a first compression spring **282a** and a second compression spring **282b** that each extend beyond the clamshell half **16a** and capable of spanning across the joint **268** between the clamshell halves **16a**, **16b**. As shown in FIG. **12B**, the first spring **282a** is seated within the clamshell half **16a** against the wire **274a** (e.g., via a wire terminal, as shown) and the second spring **282b** is seated within the clamshell half **16a** against the wire **274b** (e.g., via a wire terminal, as shown). In this embodiment, the springs **282a**, **282b** are parallel and have the same diameter and length, but in other embodiments, the springs **282a**, **282b** may alternatively be coaxial with different diameters and lengths.

With reference to FIG. **13**, the wiring bridge **266** electrically connects wires **274a** and **278a** together and separately electrically connects wires **274b** and **278b** together. Specifically, compression spring **282a** is compressed between and electrically connects wires **274a** and **278a**, while compression spring **282b** is compressed between and electrically connects wires **274b** and **278b** as the clamshell halves **16a**, **16b** are coupled together. The compression springs **282a**, **282b** serve as electrical conductors between the first set of wires **274a**, **274b** and the second set of wires **278a**, **278b**, such that the wiring bridge **266** is composed of an electrically conductive material. When the clamshell halves **16a**, **16b** are coupled, the compression springs **282a**, **282b** are compressed against wires **274a**, **274b**, **278a**, **278b**, respectively, to ensure that contact is maintained, and electrical connection is not inadvertently interrupted. Although the wiring bridge **266** of the illustrated embodiment is comprised of a set of compression springs, in other embodiments, the wiring bridge **266** may alternatively be comprised of a set of leaf springs or other type of elastic, conductive bodies.

With reference to FIG. **17**, the platen **118** includes a front edge **350** adjacent the driven wheel **114**, a rear edge **354** adjacent the drive wheel **110**, a bottom side **358** that is configured to be in a facing relationship to a workpiece, and a top side **362** that is disposed opposite the bottom side **358**. The top side **362**, in particular, is in a facing relationship to

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the motor **34**. The platen attachment **120** is removably coupled to the platen **118** and extends between the front and rear edges **350**, **354** on the bottom side **358** of the platen **118**. In the illustrated embodiment, a nitride surface coating exists on the platen attachment **120** to decrease the coefficient of friction between the platen attachment **120** and the sanding belt **30**. In some embodiments, the platen attachment **120** is composed of 1080 steel.

With reference to FIGS. **18-21**, the platen attachment **120** includes a first lip **366** that is configured to bend around the front edge **350** of the platen **118**. That is, the platen attachment **120** extends along the bottom side **358** of the platen **118** and curls around the front edge **350** to the top side **362** of the platen **118**. The platen attachment **120** includes a plurality of slots **370** disposed along the first lip **366** that are configured to correspondingly receive a plurality of pins **374** disposed on the top side **362** of the platen **118**. The plurality of slots **370** align with the plurality of pins **374**, which are evenly spaced apart along the top side **362** of the platen **118** in a direction parallel to the driven wheel axis **130**. The plurality of pins **374** are not accessible by a user unless the belt sander **10** is disassembled. The plurality of slots **370** are L-shaped such that one leg of the slots **370** extends along a direction perpendicular to the driven wheel axis **130** and another leg of the slots **370** extends along a direction parallel to the driven wheel axis **130**. The L-shape of the slots **370** assists with assembly of the platen attachment **120**, as described in further detail below. In the illustrated embodiment, there are four slots **370** and four pins **374**, while in other embodiments, there may alternatively be more or fewer than four slots **370** and corresponding pins **374**. Although the illustrated embodiment incorporates the plurality of pins **374**, in other embodiment, the pins **374** may alternatively be fasteners, screws, or the like threaded into the platen **118**.

With continued reference to FIGS. **18-21**, the platen attachment **120** further includes a tab **378** disposed on the first lip **366**. The tab **378** extends downward from the top side **362** toward the bottom side **358** of the platen **118**. The tab **378** is configured to receive a fastener **382** (FIG. **21**) that threads into a sidewall **384** of the platen **118** to further couple the platen attachment **120** to the platen **118**, as described in further detail below.

With reference to FIG. **22**, the platen attachment **120** further includes a second lip **386** that is disposed adjacent the bottom side **358** and the rear edge **354** of the platen **118**. In the illustrated embodiment, the second lip **386** bends along a corresponding bend of rear edge **354** of the platen **118**. The second lip **386** bends along a large radius and extends along a path that is tangential to the radius of the drive wheel **110**. As such, the sanding belt **30** and the second lip **386** extend along parallel paths when the sanding belt **30** no longer engages a workpiece. The second lip **386** decreases wear rates of the sanding belt **30** and decreases temperature generation between the platen attachment **120** and the sanding belt **30**.

FIG. **23** illustrates a second lip **386'** in accordance with another embodiment of the platen attachment **120**. Here, the second lip **386'** bends along a continuous arc defining a constant radius. Specifically, the second lip **386'** extends beyond the rear edge **354** of the platen **118** and bends above at least a portion of the bottom side **358** of the platen **118**. A single line may extend between the second lip **386'** and the drive wheel **110** that is tangential to both the second lip **386'** and the drive wheel **110**. The second lip **386'** decreases wear

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rates of the sanding belt 30 and decreases temperature generation between the platen attachment 120 and the sanding belt 30.

During assembly of the platen attachment 120 to the platen 118, a user simply engages the platen attachment 120 with the bottom side 358 of the platen 118, where the first lip 366 is disposed forward of the front edge 350, as shown in FIG. 19. At this point, the user aligns the plurality of slots 370 with the plurality of pins 374 and slides the platen attachment 120 rearward towards the drive wheel 110 (FIG. 19). As a result, the plurality of pins 374 are received within the leg of slots 370 that extends perpendicular to the driven wheel axis 130 (FIG. 20). Subsequently, a user slides the platen attachment 120 in a direction parallel to the drive wheel axis 130, such that the plurality of pins 374 are received within the leg of the slots 370 that extends parallel to the driven wheel axis 130. Now, the tab 378 abuts the sidewall 384 of the platen 118, where the fastener 382 can be threaded into the sidewall 384 of the platen 118 to rigidly couple the platen attachment 120 to the platen 118. In other words, the platen attachment 120 is inhibited from moving relative to the platen 118, until the fastener 382 is removed.

With reference to FIGS. 24 and 25, the belt sander 10 may further include a wear skid 390 removably coupled to the rear edge 354 of the platen 118. Specifically, the wear skid 390 protrudes beyond the platen attachment 120 and extends into the path of the sanding belt 30 to provide a more gradual transition to the sanding belt 30 between the platen attachment 120 and the drive wheel 110. As a result, the sanding belt 30 is prevented from rubbing against the rear edge 354 of the platen 118, which may otherwise occur in absence of the wear skid 390, which may cause premature wear of the sanding belt 30.

With continued reference to FIGS. 24 and 25, the wear skid 390 defines a continuous arc defining a constant radius and extends across the entire width of the platen 118. The wear skid 390 is coupled to an enlarged head 394 of the platen 118. The enlarged head 394 includes a notch 398 to facilitate assembly of the wear skid 390 onto the enlarged head 394 and facilitate heat transfer away from the wear skid 390. In some embodiments, the wear skid 390 is composed of 1080 steel and includes the nitride surface coating to decrease the coefficient of friction between the wear skid 390 and the sanding belt 30. In other embodiments, the wear skid 390 may alternatively be composed of a ceramic material. Although not illustrated, the wear skid 390 may alternatively be a rolling element that rotates in response to engagement with the sanding belt 30 as the sanding belt 30 is being driven.

Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

Various features and advantages of the invention are set forth in the following claims.

What is claimed is:

1. A belt sander for sanding a workpiece and defining a center of gravity within a first plane, the belt sander comprising:

- a main housing;
- a handle defining a second plane;
- a drive unit disposed within the main housing, the drive unit including
 - a motor disposed adjacent a first lateral sidewall of the main housing,

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a pulley system disposed adjacent a second lateral sidewall of the main housing opposite the first lateral sidewall, and

a planetary gear transmission disposed between the motor and the pulley system; and

a belt drive system driven by the drive unit, the belt drive system including

a drive wheel that is driven by the pulley system,

a driven wheel that is driven by the drive wheel via a sanding belt, and

a platen disposed between the drive wheel and the driven wheel to press the sanding belt against a workpiece,

wherein the platen defines a third plane extending perpendicular therefrom,

wherein the planetary gear transmission and pulley system are disposed on one side of the first plane and the motor is disposed on an opposite side of the first plane, and

wherein the motor is intersected by the second plane and the third plane, and a drive shaft that provides a rotational input to the planetary gear transmission, and an output shaft that is driven by the planetary gear transmission and provides a rotational input to the pulley system, wherein the drive shaft, the planetary gear transmission, and the output shaft are coaxial.

2. The belt sander of claim 1, wherein the planetary gear transmission is disposed adjacent the second lateral sidewall.

3. The belt sander of claim 1, wherein the first plane is disposed between the motor and the planetary gear transmission.

4. The belt sander of claim 1, wherein the first plane is approximately equal distance from the first lateral sidewall and the second lateral sidewall.

5. The belt sander of claim 1, wherein the first plane, the second plane, and the third plane are all parallel to each other and perpendicular relative to the platen.

6. The belt sander of claim 1, wherein the main housing includes a first clamshell half, a second clamshell half, and a joint along which the first and second clamshell halves are joined, and wherein the second plane is coplanar with the joint.

7. The belt sander of claim 1, wherein the first plane is closer to the second plane than the third plane.

8. The belt sander of claim 1, wherein the handle is a first handle, and wherein the belt sander further includes a second handle movably coupled to the main housing and a lock system to selectively secure the second handle relative to the main housing.

9. The belt sander of claim 8, wherein the lock system is pivotable between a locked state, in which the second handle is inhibited from moving relative to the main housing, and an unlocked state, in which the second handle is permitted to move relative to the main housing.

10. The belt sander of claim 8, wherein the main housing includes a rail along which the second handle is slidable along a rail axis.

11. The belt sander of claim 10, wherein the second handle is lockable to the rail in two discrete positions.

12. The belt sander of claim 10, wherein the lock system includes an axle received within the second handle, a lever pivotable about the axle to move the lock system between a locked state and an unlocked state, and a cam body coupled to and rotatable by pivoting the lever.

13. The belt sander of claim 12, wherein the cam body is received within a recess in the rail when the lock system is

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in the locked state, and wherein the cam body is removed from the recess when the lock system is in the unlocked state.

14. The belt sander of claim 13, wherein the cam body mechanically interferes with the recess when the lock system is in the locked state, such that the second handle displaces in a direction perpendicular to the rail axis to maintain frictional contact between a track of the second handle and the rail of the main housing.

15. The belt sander of claim 14, wherein the lock system further includes an elastomeric bushing disposed around the axle, which is compressed when the second handle displaces in a direction perpendicular to the rail axis.

16. The belt sander of claim 12, wherein the lever includes a tip disposed opposite from the cam body that extends beyond the second handle to allow a user to grasp the lever.

17. A belt sander for sanding a workpiece and defining a center of gravity within a first plane, the belt sander comprising:

- a main housing;
- a first handle defining a second plane;
- a second handle movably coupled to the main housing and a lock system to selectively secure the second handle relative to the main housing, the lock system including an axle received within the second handle,
- a lever pivotable about the axle to move the lock system between a locked state and an unlocked state, and a cam body coupled to and rotatable by pivoting the lever;

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a rail disposed on the main housing along which the second handle is slidable along a rail axis;

a drive unit disposed within the main housing, the drive unit including

a motor disposed adjacent a first lateral sidewall of the main housing,

a pulley system disposed adjacent a second lateral sidewall of the main housing opposite the first lateral sidewall, and

a transmission disposed between the motor and the pulley system; and

a belt drive system driven by the drive unit, the belt drive system including

a drive wheel that is driven by the pulley system,

a driven wheel that is driven by the drive wheel via a sanding belt, and

a platen disposed between the drive wheel and the driven wheel to press the sanding belt against a workpiece,

wherein the platen defines a third plane extending perpendicular therefrom,

wherein the transmission and pulley system are disposed on one side of the first plane and the motor is disposed on an opposite side of the first plane, and

wherein the motor is intersected by the second plane and the third plane.

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